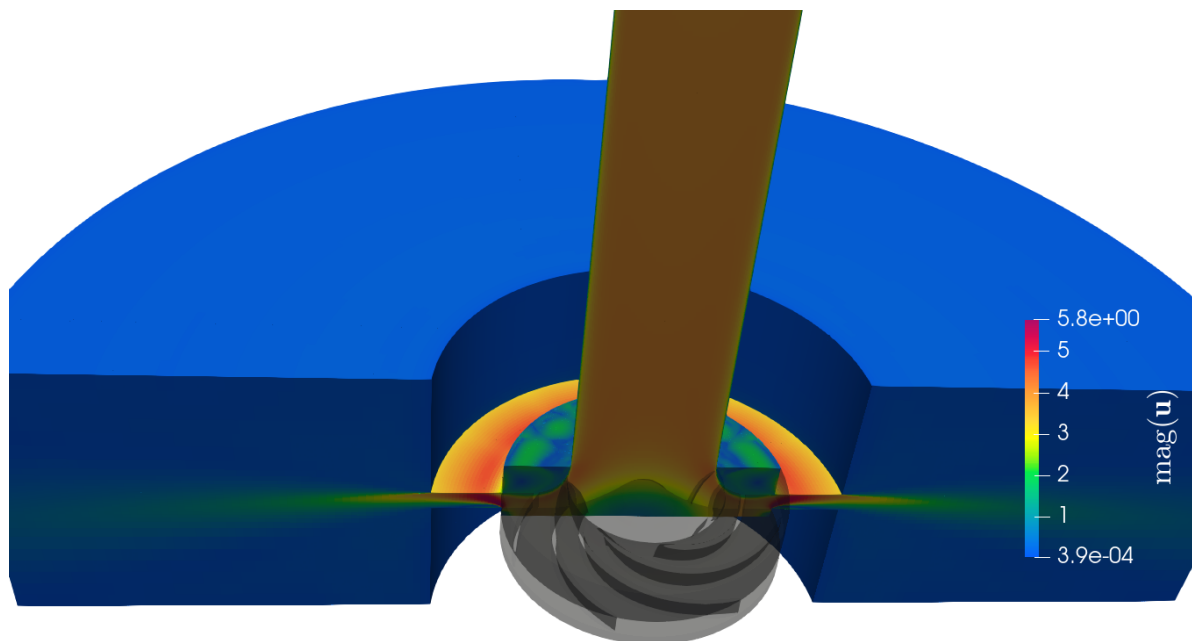


End-of-study thesis for engineering degree

Optimization of the performance of a centrifugal pump using Data Assimilation

Intern in Rotating Machines



Cross-section of the pump model

Educational tutor: Samy LABSIR

Laboratory tutors: Marcello MELDI, Antoine DAZIN, Miguel MARTINEZ VALERO

Clément THIBAULT - Aéro 5

From February 26th, 2025 to September 2nd, 2025

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Summary sheet

The aim of this technical assessment is to summarize the gaps and their causes between the course's missions and what has been achieved. It also enables us to suggest what could be done by the next person working on the same subject.

Summary sheet		Clément THIBAULT - Aéro 5 TIE	
Internship topic		Objectives	
Optimization of the performance of a centrifugal pump using Data Assimilation		Optimize the forcing parameters of an IBM ¹ in a centrifugal pump using DA with CONES, to find the correct performance of the pump.	
Main customer		Used tools	
- LMFL - ENSAM		OpenFoam 9, CONES, VSCode, LaTeX, Git, Python,C++, MPI, Bash, TGCC, Paraview	
Conducted studies			
- Coding the DA test case in OpenFoam and CONES - Convergence of the CFD solution (physical and numerical stability/precision) - Convergence of the EnKF, adding tools as inflation			
Results		Explanation of possible differences	
- Drop up to 50% of the mathematical NRMSE with DA (almost 100% expected) - Implementation of inflation, in CONES - Implementation of radial expansion, "azimuthal interpolation" in OpenFoam 9 - Discovering of a recirculation bubble in the diffuser		- After the diffuser, the impeller's wake direction is reversed because of the MRF method - The rotating impeller shape combined with the IBM and MRF allows non-physical flow that induce a recirculation bubble. - Results of the EnKF are not as good as expected because of the IBM method combined to MRF method, which is not adapted to the centrifugal pump geometry.	
Encountered difficulties		Work to be continued	
- Understanding the tools OpenFOAM and CONES - Code everything directly on the TGCC terminal - Initialize physical field of complex instationary simulation		- Use the same type of DA for the analysis of axial compressor internal flows - Synchronize PIV and simulation in the DA - Implement interpolation IBM method	

Note: Acronyms are define later.

¹All acronyms are defined from the next page or available in the glossary.

General Introduction

Since I was young, I've always been fascinated by fluid mechanics. I remember watching planes flying and being amazed by the way they could fly. I was also interested in the way water flows in rivers and sailing boats. In high school, I began working on plane-related projects and explored them more theoretically through Computation Fluid Dynamics (CFD).

This fascination led me to study physics and mathematics at university, where I discovered applied mathematics including CFD, Artificial Intelligence (AI) and Kalman Filter. During my studies, I've become passionate about applied mathematics and fluid mechanics. That's why I did my last internship at the "Centre Européen de Recherche et de Formation Avancée en Calcul Scientifique" (CERFACS). It convinced me to do a thesis in CFD after "Institut Polytechnique des Sciences Avancées" (IPSA), my current engineering school.

The subject proposed by the "Laboratoire de Mécanique des Fluides de Lille" (LMFL) really suited me: CFD using an open-source library, coupled with AI and experimental data. So it was with enthusiasm that I started this internship on February 26, 2025, in this laboratory for a duration of six months.

Briefly, the LMFL is a public laboratory, consisting of 38 researchers and around 25 PhD students. The laboratory is present on three sites located in Lille ("École Nationale Supérieure d'Arts et Métiers" (ENSAM) and "Office National d'Études et de Recherche Aéronautiques" (ONERA)) and on the Villeneuve d'Ascq campus (Centrale Lille, "Centre National de la Recherche Scientifique" (CNRS) and University of Lille).

My main tasks during this internship were:

- Ensuring the convergence of the CFD solution;
- implementing the Data Assimilation (DA) problem in Open source Field Operation And Manipulation (OpenFOAM) and Coupling OpenFoam with Numerical Environments (CONES);
- optimizing the forcing parameters of the Immersed Boundary Methods (IBM) using DA with CONES to correct a non-physical behavior.

To present the details of my internship work, I will first introduce the LMFL research laboratory. In the second part, I will describe the work I carried out, structured into the following six sections:

1. State of the art;
2. Numerical ingredients;
3. Experimental data and pre-processing;
4. IBM preliminary results;
5. Application of the DA;
6. results

Chapter 1

The LMFL laboratory and Arts et Métiers Lille campus

1.1 Scientific environment and host institution

The LMFL is a public research laboratory jointly supervised by Centrale Lille, Université de Lille, Arts et Métiers ParisTech, ONERA, and the CNRS.



Figure 1.1: LMFL facilities; Top: Villeneuve d'Ascq campus, bringing together Centrale Lille, the CNRS, and Université de Lille. Bottom-left: ONERA. Bottom-right: Arts et Métiers campus (ENSAM) (source : lmfl.cnrs.fr)

Over the past six months, I have been hosted by ENSAM, specifically at the Lille campus. Arts et Métiers is a prestigious engineering school focused on mechanical, industrial, and energy engineering. The Lille campus gives to students a dynamic and innovative environment in a region with a strong industrial heritage.

The LMFL was created on January 1, 2018, from the merger of two research units: the “Écoulements Tournants et Turbulents” team from the Lille Mechanics Laboratory, and the “Expérimentation et Limite de Vol” unit. This diversity of expertise enables the research teams to engage in numerous collaborative projects with regional, national and international partners.

Research conducted within the LMFL is organized into three main topics:

- **Turbulence:** Studying and modeling turbulent flows using numerical and experimental methods. The focus is on inhomogeneity and non-stationarity in wall-bounded flows, wakes, jets, and various other flow configurations. This research area includes ten permanent researchers, as well as an equal number of PhD students and postdoctoral researchers.
- **Rotating flows:** Experimental and numerical studies of rotating machines (turbomachines, rotors, propellers) operating under degraded conditions, as well as flow control, cavitation, and multiphase flows, are also key topics. This subject gathers six researchers and about ten PhD and postdoctoral students.
- **Flight dynamics in unsteady and non-uniform environments:** Development of analytical and experimental methods to determine the evolving behaviour of low-speed aircraft in perturbed flight domains. Post-stall dynamics, environmental interactions, and experimental design are among the main research interests. This topic involves about five research engineers and ten PhD and post-doctoral students.

1.2 Social and environmental responsibility

ENSAM's Corporate Social Responsibility (CSR) (or "Responsabilité Sociétale des Entreprises" (RSE)) action plan is structured around five strategic axes¹:

1. **Strategy, governance and territorial anchoring:** presents ENSAM's actions in terms of sustainable development, in particular the Evolutive Learning Factories 4.0 project aimed at modernizing training and research platforms by fully integrating the CSR dimension;
2. **Education and training:** Training students in eco-responsibility;
3. **Research and innovation:** Ensure that all research themes have direct or indirect impacts on the achievement of international and national sustainable development goals. Ensure free access to research results and data;
4. **Environmental management:** ENSAM aims to reduce its environmental impact by improving energy efficiency, cutting emissions (including Scope 3), and integrating carbon accounting. It also promotes biodiversity and sustainable mobility through staff engagement, responsible purchasing, and green-space management;
5. **Social policy:** Reconciling economic, societal, and environmental dimensions by training staff, teachers, research professors, and stakeholders in sustainable development issues, developing a quality of life policy within the institution, and ensuring equal opportunities in training, research, and human resources. An indicator is the "Qualité de Vie au Travail" (QVT), that is detailed in the "Rapport Social Unique" (RSU)² 2024.

The "Comité Social et Économique" (CSE) is the staff representative body within ENSAM. Although specific information on the CSE is limited to the 2022 RSU mentioning the monitoring of the actions of the "Comité d'Hygiène, de Sécurité et des Conditions de Travail National" (CHSCTN) which is the old name of the CSE.

¹<https://artsetmetiers.fr/fr/ecole-engagee>, https://dp-www.s3.ensam.eu/public/2024-11/24_032_plaquette-bilan-RSE_V9_0.pdf

²https://dp-www.s3.ensam.eu/public/2025-07/RSU%202024_VCA.pdf

Arts et Métiers also publishes its activity reports, its social balance sheet / single social report and public data then transmitted to the "Commission des Titres d'Ingénieur" (CTI) available online ³. More specifically, students from Arts et Métiers created a webpage to learn about general CSR and sustainable development: ⁴ (addressing themes such as ecology, inclusion, gender equality and the fight against discrimination,...).

About the LMFL we can also find an interesting documents from "Haut Conseil de l'Évaluation de la Recherche et de l'Enseignement Supérieur" (HCERES). This organization is in charge of evaluating all research structures. Every five years, it publishes a report on the LMFL. The last one⁵ was published in February of this year. It presents the laboratory's strengths and weaknesses, as well as its research activities, collaborations, workforce, and future perspectives. It highlights the laboratory's commitment to excellence in research, its multidisciplinary approach, and its contributions to the field of fluid mechanics and related areas. More precisely, the report emphasizes the laboratory's strengths in experimental and numerical methods, its expertise in turbulence and rotating flows, and its contributions to the understanding of complex fluid phenomena.

Some key takeaways from the report:

- The facilities are unique, with wind tunnels and turbomachinery test rigs that allow advanced experimental investigations of external and internal flows.
- It is one of the most active European laboratories in the field of next-generation aircraft design, particularly on electric propulsion systems, supported by high-level experimental setups such as the Eiffel wind tunnel (L1) and a vertical wind tunnel with a rotating balance. The LMFL also leads European research on airship aerodynamics and urban drone flight.
- Its experimental platforms are completed by advanced numerical simulations, with equipment partly acquired through regional "Contrat de Plan État-Région" (CPER) programs. The lab conducts original research on turbulence dissipation mechanisms in non-homogeneous, unsteady flows and on turbulent/non-turbulent interfaces.
- Efforts have been made to create synergy between the lab's three geographically separated sites through transversal research themes in optical metrology, flow control, and data analysis. Weekly LMFL webinars support internal collaboration.
- Despite some staff departures, LMFL has maintained its attractiveness by recruiting top researchers, strengthening historical themes like turbulence and instabilities, and expanding into areas like two-phase aerodynamics and data-driven modeling.
- LMFL's scientific output is of high quality, with publications in top journals (e.g., Journal of Fluid Mechanics (JFM)) and participation in major conferences. While flight dynamics publications are fewer, this area is expected to grow through collaborations, particularly with the turbulence team.
- Finally, the lab brings in solid external funding from ERC, ANR, Clean Sky, etc., and works closely with industry partners like CNES and Ariane Group. The leadership was praised for keeping the lab well-organized and for encouraging collaboration across research themes.

³<https://artsetmetiers.fr/fr/observatoire-des-donnees>

⁴<https://artsetmetiers.fr/fr/actualites/developpement-durable-et-rse-faites-le-plein-de-connaissances-sur-savoir>

⁵https://www.hceres.fr/sites/default/files/media/publications/rapports_evaluations/pdf/E2026-EV-0590349J-DER-ER-DER-PUR260025134-ST5-LMFL-RF.pdf

My personal experience of the work environment has been very positive. The working atmosphere within the laboratory is pleasant and conducive to research. The researchers are passionate about their work and share their knowledge with enthusiasm. The team is close-knit and it collaborates with researchers from all over the world. The working atmosphere suits me: researchers generally work hard, but there is no pressure regarding working hours. The building is a beautiful old, typical Lille building, but well renovated and adapted to the research work being conducted (both numerical and experimental). The laboratory offers spacious areas, interior courtyards, large lecture halls, and a large workshop for experiments.

Chapter 2

Optimizing the performance of a centrifugal pump with data assimilation

To understand fluid behaviour, scientists rely on physical measurements and governing equations. While analytical solutions exist for simple cases, more complex flows require numerical simulations. However, high-fidelity simulations remain too computationally expensive for industrial use. This limitation has led to the development of approximate models involving tunable parameters. These parameters are calibrated for specific flow conditions, using experimental data. Techniques such as the Immersed Boundary Methods (IBM), which will be detailed later, use a global approach, while the physics of the flow can locally be very different. That's why it relies on such high-fidelity data calibration. Manual tuning is feasible for a small number of parameters but becomes inefficient and inaccurate as complexity increases. This justifies the use of mathematical approaches like the EnKF, which motivated the definition of my internship topic.

The internship began in February 2025. I was provided with a fix computer, credentials to connect to the Irene and Cassiopée supercomputers, and the link to the Coupling OpenFoam with Numerical EnvironmentS (CONES)'s gitlab¹. Marcello MELDI, Antoine DAZIN and Miguel MARTINEZ VALERO who are my internship supervisors, showed me around the lab. Then, Tom MOUSSIE and Miguel MARTINEZ VALERO, my co-offices, explained me the basics of Open source Field Operation And Manipulation (OpenFOAM) and CONES, the DA code. I was able to run a few tests on OpenFOAM on the supercomputers, which allowed me to familiarize myself with the tools and the code.

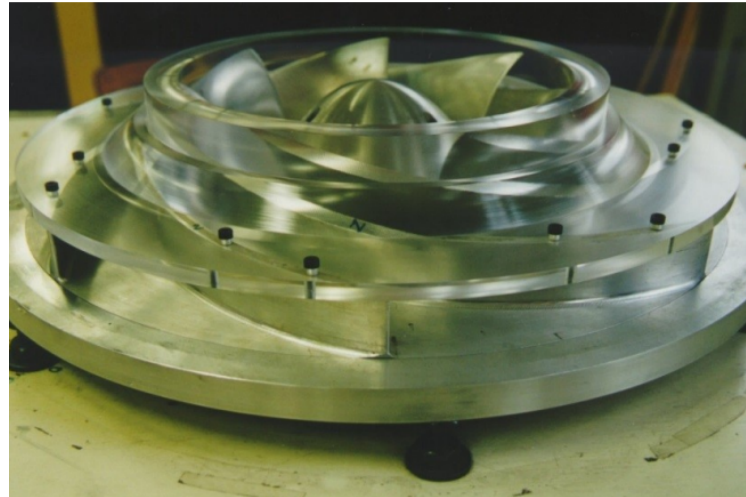
¹<https://gitlab.ensam.eu/pe431/cones-dev>

2.1 State of the art

The pump Centrifugal pumps are used since 1475[11] and are still widely used today in various applications, such as water supply, irrigation, turbomachines, and industrial processes. They are known for their low cost, efficiency and reliability, making them a popular choice for many fluid transport applications. The studied centrifugal pump is a "Société Hydrotechnique de France" (SHF) radial impeller, which is a type of centrifugal pump designed for high flow rates and low head applications. The RESEDA benchmark, located in the lab facilities, is used to study complex flow on centrifugal machines as rotor/stator interactions, instabilities, performance degradation, pressure fluctuation, highly unsteady effects,...[12]



(a) Old centrifugal pump used to move hay (personal picture)



(b) SHF impeller used in the RESEDA benchmark

Figure 2.1: Centrifugal pumps

Different high-fidelity data has been recorded on the diffuser of the pump at the LMFL:

- High-Speed Particle Image Velocimetry (PIV) data obtained by Dazin et al. [4] in 2011;
- pressure data obtained by Fan et al. [5] in 2022.

Both experimental (traditionally) and numerical (increasingly in recent years) approaches have been used to investigate the rotor-stator interactions to improve performance, develop accurate models and permit real-time monitoring and control.

IBM To obtain a numerical simulation of the flow in the pump, a Computation Fluid Dynamics (CFD) can be performed. There are different ways to discretize the space when we do CFD (see Figure 2.2²). The most common are structured meshes, unstructured meshes and body-fitted meshes.

²https://www.solidworks.com/sw/docs/flow_basis_of_cad_embedded_cfd_whitepaper.pdf

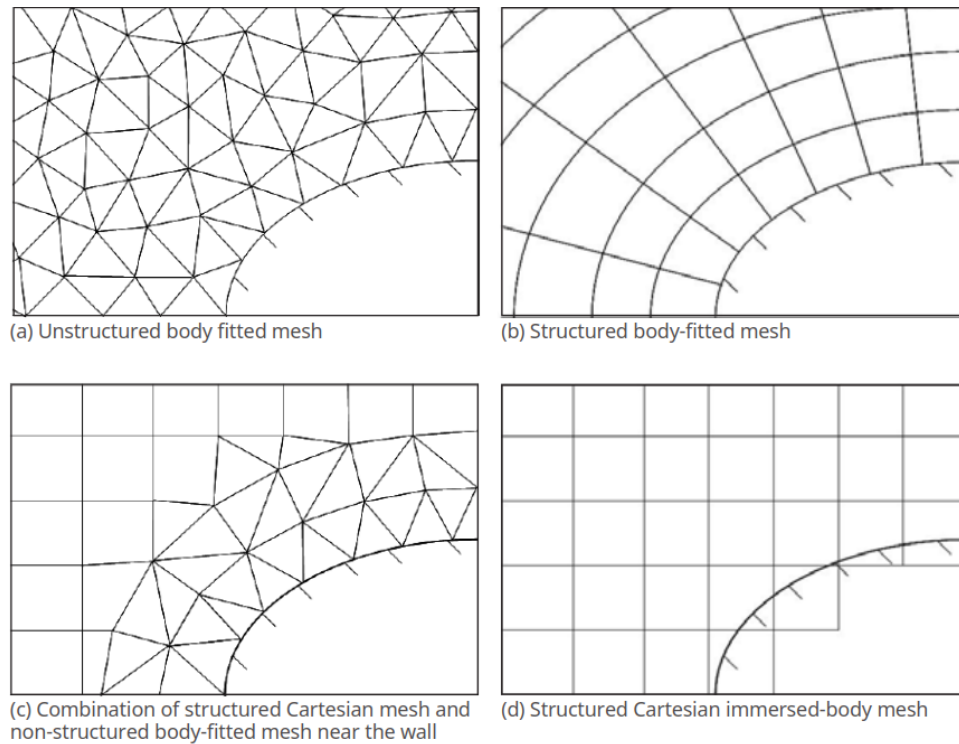


Figure 2.2: Different strategies for meshing

A structured mesh is a mesh whose connectivity between points can be described in a regular manner, typically using multidimensional arrays (1D, 2D, 3D). In such a mesh, the points are organized in a regular grid, allowing easy navigation through the data. In contrast, an unstructured mesh is a mesh where cells cannot be described in a regular manner. They are defined by arbitrary geometric shapes where the connectivity of the points does not follow a regular pattern. These meshes are often used to model complex geometries where a regular grid would be unsuitable.

The body-fitted meshes are the most accurate but also the most expensive in terms of computational resources in case the grids update at each time step.

IBM is a numerical technique that simulates viscous flow with embedded boundaries on grids that do not conform to those boundaries. This allows to keep numerical efficiency on complex geometries and moving boundaries without the need for body-fitted meshes, which can be computationally expensive and difficult to implement.

In 1972 Charles S. Peskin[9] developed the IBM method. His goal was to reduce the computational time, mainly due to the computation of the mesh refinement based on the Courant-Friedrichs-Lewy (CFL). He added an IBM forcing term on Navier-Stokes equations, only on the simulated artificial heart valve part. He overcomes the numerical instabilities due to the boundary forces by using an iterative method for solving a fixed-point problem for boundary forces.

In 1999 [2] discovered that using the penalization IMB, the velocity penalized at the wall as the same as the one from a body-fitted approach if " D ", a forcing parameter define in the subsection 2.2.1, tend to infinity.

IBM are usually less accurate and need more cells element to keep good precision in the near-wall region, but more efficient in terms of computational resources when the geometrie is moving. So it's more and more used in CFD [15]. The maths behind the method are detail in the Equation 2.23.

Data Assimilation Data Assimilation (DA) is a technique used to improve the accuracy of numerical models by combining them with observational data[3]. It is widely used in various fields, including meteorology, oceanography, and engineering. The goal of DA is to obtain a more accurate representation of the state of a system by integrating information from different physical measurements or high-fidelity sources.

There exist many methods for data assimilation, distributed in three main categories: **variational methods**, **statistical methods** and **hybrid** (see Figure 2.3).

- Variational methods (also called classical methods), such as the 3D-Var and 4D-Var, use optimization techniques to find the best estimate of the state of a system by minimizing the difference between model predictions and observations[10];
- Statistical methods (also called sequential methods), such as the Kalman filter and its Ensemble Kalman Filter (EnKF) variant, update the model state iteratively as new observations become available[3]. It is working by maximizing a certain Probability Density Function (PDF) $\langle e^a \rangle_k = 0$;
- And Hybrid methods, combining both variational and statistical that we are not going to explain here.

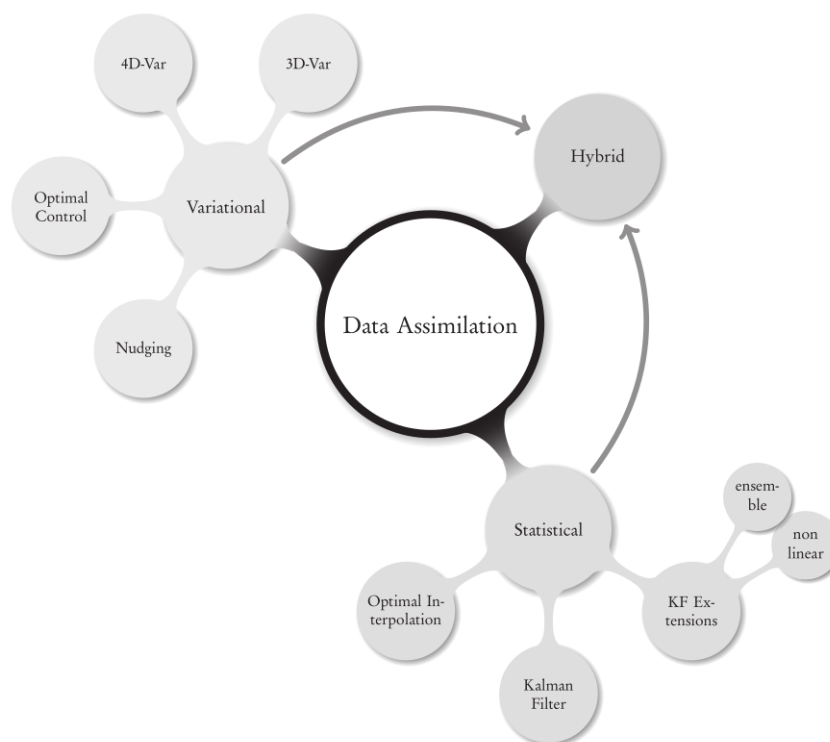


Figure 2.3: The big picture for DA methods and algorithms[3]

EnKF The EnKF, is statistica Monte Carlo DA method that uses a set of model states (referred to as ensemble member) to estimate the state of the system. The EnKF is particularly well-suited tool for assimilating data on nonlinear and high-dimensional systems and has been successfully applied in various fields, including CFD simulations, allowing for the correction of model predictions based on observed data. It mainly consists of two steps:

- **Forecast**: time advancement of the different model states or ensemble members;
- **analysis**: correction of the model prediction by some measured observation describing the same physical phenomenon.

The maths behind the EnKF method are detailed in the subsection 2.2.3.

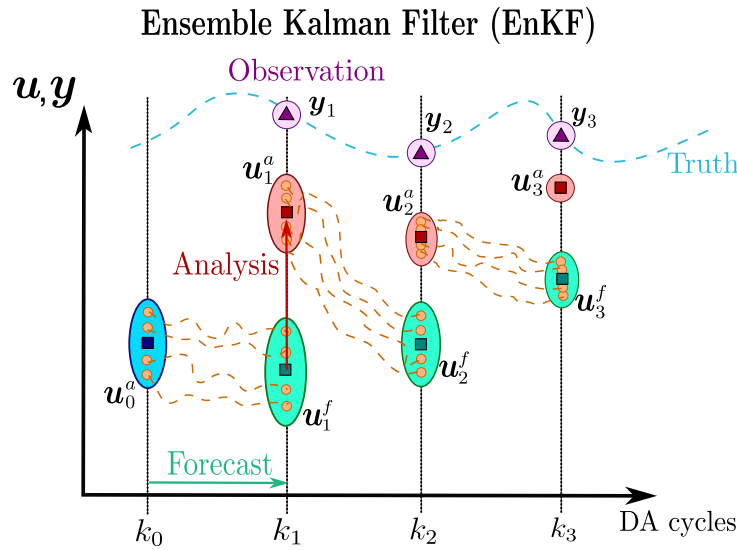


Figure 2.4: Illustration of the EnKF process (100% confidence in observations)

EnKF for DA in CFD has been used at "Laboratoire de Mécanique des Fluides de Lille" (LMFL) for a few years now.

- In 2023, Lucas Villanueva, with Miguel M.Valero and Marcello Meldi from LMFL wrote an article[16] about the amelioration of the flow around a building preforming EnKF on five CFD parameters, showing a global improvement for both the velocity field and the pressure field;
- In 2024, Tom Moussie, with Paolo Errante and Marcello Meldi from LMFL wrote an article[8] successfully optimizes the boudary condition of the academical rectangular 5:1 cylinder benchmarck using EnKF, leading to a significant improvement of the accuracy in the prediction of the turbulent flow;
- During my internship, Tom Moussie submit an article on the determination of the incidence angle of a 2D wing using EnKF, and Miguel submit also an article about the optimization of forcing parameters of an IBM around an 2D oscillating cylinder, showing very good results.

This led us to the idea of using the EnKF method to optimize the parameters of the IBM forcing term on a centrifugal pump, which is a more complex case than the previous ones. It combines 3D turbulent boundary layer, such as the turbulent plane channel flow and a moving body susch as the oscillating cylinder. Also, it's complex shape induce 11 million (11M) cells. Finally, the rotor/stator interaction in centrifugal machines is a very complexe procedure affected by highly unsteady flow phenomena, turbulent boundary layers and instabilities. CFD is generally not able to capture it properly with traditional physical models. The goal is the to improve the prediction of the flow on the diffuser, to capture the good performance of the pump by optimizing the parameters of the IBM forcing term, which will lead to a better simulation of the fluid flow and a better understanding of the pump's behaviour.

In this context, we would like to employ these two innovative methodologies to improve the original CFD predictive capabilities for a centrifugal pump test case, leveraging some PIV measurements at the diffuser, and the information from pressure taps at both the impeller inlet and the diffuser. To do so, we pursue the following objectives, folowing the manuscript organisation:

1. Do a state of the art about the centrifugal pumps, CFD methods including IBM and DA including EnKF, detail in this section 2.1;

2. understand some numerical ingredients as CFD governing equations, the open-source code OpenFoam, the EnKF method, the CONES library and the pump modeling that are explained in the subsection 2.2.1;
3. pre-process the experimental data to be able to compare them to the numerical model, section 2.3;
4. modify OpenFOAM to obtain first simulation results using IBM, section 2.4;
5. modify CONES to adapte it to the pump case and apply the DA to modify the IBM parameters, section 2.5;
6. analyse the results of the DA on the optimization of the field in the diffuser and the accuracy of the performance of the pump, section 2.6.

2.2 Numerical ingredients

2.2.1 CFD governing equations

Navier-Stokes equations CFD is based on the fundamental equations of fluid mechanics. When the fluid is assumed to be Newtonian, these equations reduce to the nonlinear Navier–Stokes partial differential equations.

These equations express three physical conservation principles, given here in their Eulerian form (i.e., fixed spatial reference frame), which is the standard in CFD because it describes how fluid properties evolve at fixed locations in space, which is ideal for mesh-based numerical simulations.

- **Mass conservation (Continuity equation):**

$$\frac{\partial \rho}{\partial t} + \operatorname{div}(\rho \mathbf{U}) = 0 \quad (2.1)$$

With P the pressure, ρ the density and $\mathbf{U}$ the velocity.

- **Momentum conservation (second Newton Law):**

$$\frac{\partial(\rho \mathbf{U})}{\partial t} + \nabla \cdot (\rho \mathbf{U} \otimes \mathbf{U}) = -\nabla P + \nabla \cdot \boldsymbol{\tau} + \mathbf{f} \quad (2.2)$$

where $\boldsymbol{\tau}$ is the viscous stress tensor and $\mathbf{f}$ includes body forces (e.g., gravity, source terms).

- **Energy conservation (First law of thermodynamics):**

$$\frac{\partial(\rho E)}{\partial t} + \nabla \cdot (\mathbf{U}(\rho E + P)) = \nabla \cdot (k \nabla T) + \Phi \quad (2.3)$$

where k is the thermal conductivity, T the temperature, and Φ the viscous dissipation and e the potential energy. $E = e + \frac{1}{2} \mathbf{U}^2$ is the total energy per unit mass, with:

- e : internal energy per unit mass (also called specific internal energy)
- $\frac{1}{2} \mathbf{U}^2$: kinetic energy per unit mass (specific kinetic energy)

For an ideal gas, the internal energy is generally given by $e = c_v T$, where:

- c_v is the specific heat capacity at constant volume
- T is the temperature

Two main indicators of the main behaviour of the fluid are the Mach number (Ma) and the Reynolds number (Re).

The Ma , is defined by the velocity of the fluid divided by the speed of the sound (relative to the fluid). In CFD, fluid is usually considered as incompressible for Mach lower than 0.3. In our case, in atmospheric conditions,

$$Ma = \frac{R_2 \Omega}{c} = 0.095 \quad (\text{incompressible}) \quad (2.4)$$

As we consider that we are in **incompressible** flow conditions (because we are at low Ma), Equation 2.1 becomes

$$\nabla \cdot \mathbf{u} = 0 \quad (2.5)$$

And the pressure and velocity are solved iteratively, contrary on compressibles flows.

The dimensionless Re measure the ratio between inertial and viscous forces. It indicates if the flow is turbulent, laminar or both, it characterizes the flow regime. Let's compute the Re around where the velocity should be the highest.

$$Re = \frac{UL}{\nu} \quad (2.6)$$

Where U is the velocity of the fluid, L is the characteristic length and ν the kinematic viscosity of the fluid.

$$Re_{r_{tip}} = \frac{R_2^2 \Omega}{\nu} = 5.53 \times 10^5 \quad (2.7)$$

Turbulence modelling This equations permit running all the scales in case of turbulent flows, i.e. a Direct Numerical Simulation (DNS), that would be too costly in our case. The Kolmogorov scale, defined by the spatial scale from which viscosity allows the kinetic energy of a flow to be dissipated, is very small. In fluid mechanics, turbulent flows consist of several scales which are responsible for different phenomena. As illustrated in the following Figure 2.5a³ and Figure 2.5c⁴, large scales are responsible for the production of turbulence and energy, which is shifted through the inertial range toward the Kolmogorov scale, responsible for the dissipation phenomena. There are two major ways to model the turbulence. The first one is by using Large Eddy Simulation (LES), that simulates only the small scale eddy usually in the near-wall region. The second one is the Reynolds-Averaged Navier-Stokes equations (RANS) that model every scale of eddy. The result is a steady-state and it is less precise but faster computed than LES. The Figure 2.5b⁵ illustrates well a visualization of the fields of the three type of CFD models.

////////// Attention à la figure 2.5 parce que les labels ne sont pas correctes (la DNS et la LES n'utilisent pas des modèles de turbulence et la figure c ne regarde pas vraiment la RANS que tu as fait) //////////

³<https://www.cfdsupport.com/openfoam-training-by-cfd-support/node346/>

⁴https://help.altair.com/hwcfdsolvers/acusolve/topics/acusolve/training_manual/turbulence_modeling_r.htm

⁵<https://blog.diphyx.com/comprehensive-guide-review-to-choosing-the-right-cfd-software-in-2023-features-performance-and-7ebc0623bfa6>

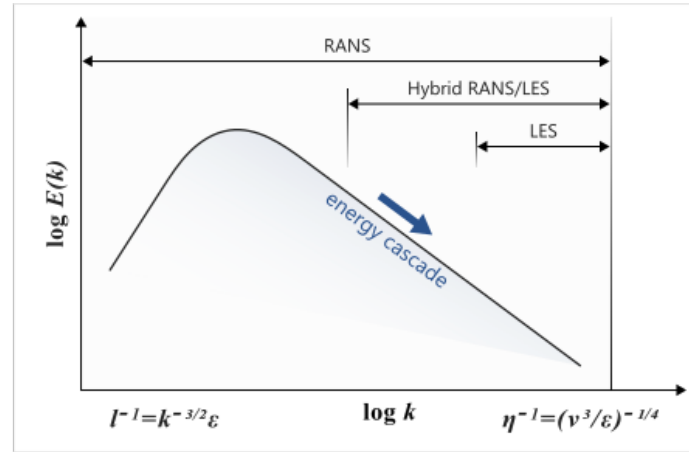
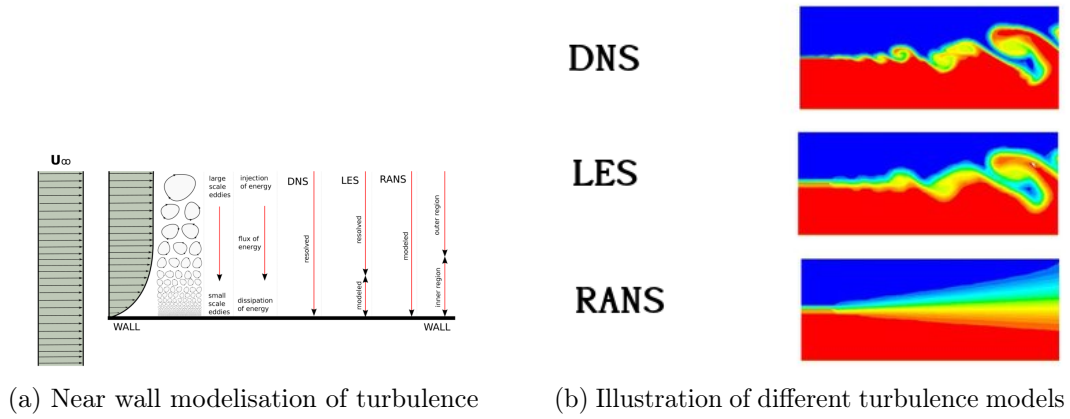


Figure 2.5: Turbulence models

In our study, we are going to have a look on the Unsteady Reynolds-Averaged Navier–Stokes (U-RANS) method use the same principle as the RANS method but it’s unsteady, allowing to make turn physically in our case.

More precisely, in our RANS and U-RANS simulation, we are going to use a $k - \omega$ Shear Stress Transport (SST) turbulence model. It is a two-equation eddy-viscosity model: it combines the $K - \epsilon$ and the $K - \omega$ turbulence approximations plus a blending region between both.

$K - \omega$ is known to be a good model in the near-wall region but a bad prediction in bulk flow. $K - \epsilon$ make a good numerical prediction in the bulk flow field but is less accurate near solid surfaces.

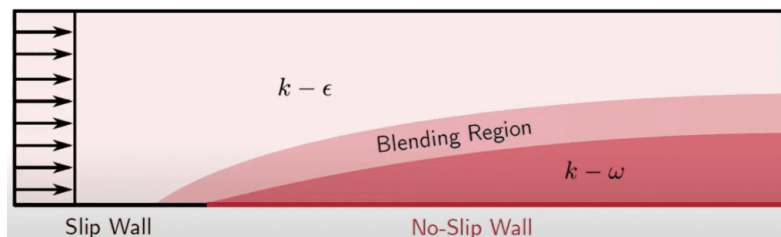


Figure 2.6: $k - \omega$ SST model

$k - \omega$ SST model is defined as follows⁶[6][7] :

⁶https://www.cfd-online.com/Wiki/SST_k-omega_model

- Norm of the strain tensor

$$S = \sqrt{2S_{ij}S_{ij}}, \quad \text{avec } S_{ij} = \frac{1}{2} \left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i} \right) \quad (2.8)$$

- Kinematic Eddy Viscosity:

$$\nu_T = \frac{a_1 k}{\max(a_1 \omega, SF_2)} \quad (2.9)$$

- Turbulent kinetic energy production term

$$P_k = \tau_{ij} \frac{\partial U_i}{\partial x_j} \quad (2.10)$$

- Turbulence Kinetic Energy :

$$\frac{\partial k}{\partial t} + U_j \frac{\partial k}{\partial x_j} = P_k - \beta^* k \omega + \frac{\partial}{\partial x_j} \left[(\nu + \sigma_k \nu_T) \frac{\partial k}{\partial x_j} \right] \quad (2.11)$$

- Specific Dissipation Rate :

$$\frac{\partial \omega}{\partial t} + U_j \frac{\partial \omega}{\partial x_j} = \alpha S^2 - \beta \omega^2 + \frac{\partial}{\partial x_j} \left[(\nu + \sigma_\omega \nu_T) \frac{\partial \omega}{\partial x_j} \right] + 2(1 - F_1) \sigma_\omega \frac{1}{\omega} \frac{\partial k}{\partial x_i} \frac{\partial \omega}{\partial x_i} \quad (2.12)$$

- Closure Coefficients and Auxiliary Relations :

$$F_2 = \tanh \left[\left[\max \left(\frac{2\sqrt{k}}{\beta^* \omega y}, \frac{500\nu}{y^2 \omega} \right) \right]^2 \right] \quad (2.13)$$

$$P_k = \min \left(\tau_{ij} \frac{\partial U_i}{\partial x_j}, 10\beta^* k \omega \right) \quad (2.14)$$

$$F_1 = \tanh \left\{ \left\{ \min \left[\max \left(\frac{\sqrt{k}}{\beta^* \omega y}, \frac{500\nu}{y^2 \omega} \right), \frac{4\sigma_\omega k}{CD_{k\omega} y^2} \right] \right\}^4 \right\} \quad (2.15)$$

$$CD_{k\omega} = \max \left(2\rho \sigma_\omega \frac{1}{\omega} \frac{\partial k}{\partial x_i} \frac{\partial \omega}{\partial x_i}, 10^{-10} \right) \quad (2.16)$$

$$\phi = \phi_1 F_1 + \phi_2 (1 - F_1)$$

$$\alpha_1 = \frac{5}{9}, \quad \alpha_2 = 0.44$$

$$\beta_1 = \frac{3}{40}, \quad \beta_2 = 0.0828$$

$$\beta^* = \frac{9}{100}$$

$$\sigma_{k1} = 0.85, \sigma_{k2} = 1$$

$$\sigma_{\omega 1} = 0.5, \sigma_{\omega 2} = 0.856$$

MRF method To simulate the rotation of the impeller in a steady-state RANS simulation, we use the Multiple Reference Frame (MRF) method. The MRF strategy consists in solving the Navier-Stokes equations in a rotating reference frame in the rotating region, while keeping the rest of the domain in a stationary (inertial) frame. The mesh and geometry remain static; only the reference frame changes.

The governing equations in the rotating frame (relative velocity) are derived from the Navier-Stokes equations in the inertial frame by applying a transformation that accounts for the rotating frame accelerations.

For an incompressible flow, the Navier-Stokes equations in the rotating frame with relative velocity $\vec{u}_R$ are:

$$\begin{cases} \frac{\partial \vec{u}_R}{\partial t} + \nabla \cdot (\vec{u}_R \otimes \vec{u}_R) + 2\vec{\Omega} \times \vec{u}_R + \vec{\Omega} \times (\vec{\Omega} \times \vec{r}) = -\nabla \left(\frac{p}{\rho} \right) + \nu \nabla^2 \vec{u}_R \\ \nabla \cdot \vec{u}_R = 0 \end{cases} \quad (2.17)$$

Where:

- $\vec{u}_R$ is the velocity in the rotating frame;
- $\vec{\Omega}$ is the angular velocity vector;
- $\vec{r}$ is the position vector;
- ρ is the fluid density;
- ν is the kinematic viscosity.

The additional terms $2\vec{\Omega} \times \vec{u}_R$ (Coriolis force) and $\vec{\Omega} \times (\vec{\Omega} \times \vec{r})$ (centrifugal force) appear due to the non-inertial frame transformation.

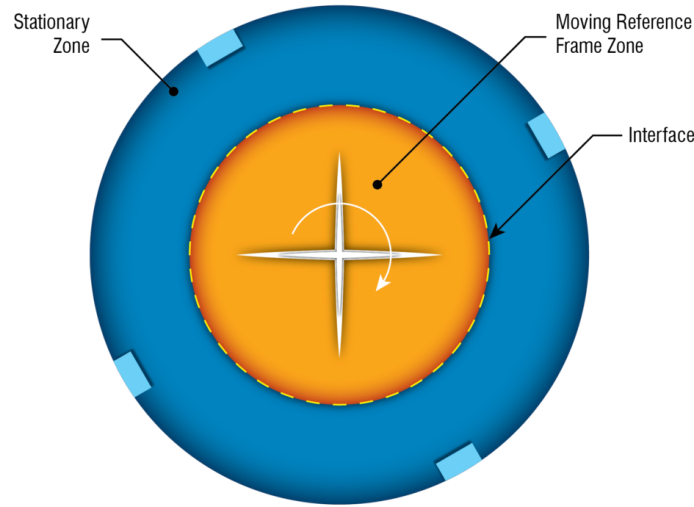


Figure 2.7: MRF strategy: the rotating region is solved in a non-inertial frame

U-RANS In the U-RANS case, the impeller is physically moving.

To do so, the CFL or the courant number has to be computed.

$$CFL = \frac{\Delta_t}{\Delta_x} \times \mathbf{U} \quad (2.18)$$

Typically set to a maximum of 1 for stability. A way to visualize the CFL is to think of it as the fraction of the distance a wave travels in one time step relative to the distance between two cells in the azimuthal direction at the tip of the impeller. The CFL is computed locally at each mesh element. Let computes the Δ_t at the tip of the impeller, where the velocity is the highest and Δ_x the lowest.

- Δ_x is the distance between two cells in the azimuthal direction at the tip of the impeller in our model, which is given as 1.5 mm or 0.0015 m.

- $\|U_\infty\|$ is the magnitude of the velocity at infinity, which can be approximated by the tangential velocity at the tip of the impeller.

$$\|U_\infty\| = \Omega \cdot r_{tip} \quad (2.19)$$

where:

- ω is the angular velocity in radians per second,

- r_{tip} is the radius at the tip of the impeller.

Assuming the radius at the tip of the impeller is $r = 0.2566$ m, we can compute the tangential velocity:

$$\|U_\infty\| = 125.66 \text{ rps} \cdot 0.2566 \text{ m} = 32.24 \text{ m/s} \quad (2.20)$$

Assuming the distance between two cells in the azimuthal direction at the tip of the impeller is $\Delta_x = 0.0015$ m (which is 1.5 mm), we can now compute the time step Δ_t :

$$\Delta_t = \frac{0.5 \cdot 0.0015 \text{ m}}{32.24 \text{ m/s}} = 2.3 \times 10^{-5} \text{ s} \quad (2.21)$$

IBM equations As see in the section 2.1, different techniques exist to penalise the immersed part. In our case, we are going to use a classical penalisation method proposed by Angot et al.[2] which enters the family of continuous forcing approach. Instead of use body-fitted mesh, we are going to consider porous medium and adding a forcing term on the Navier-Stokes equations (Equation 2.24) to simulate the effect of the immersed boundary on the fluid flow. This term is added only on the penalization part $\Omega_b U \Sigma_b$ (see Figure 2.8).

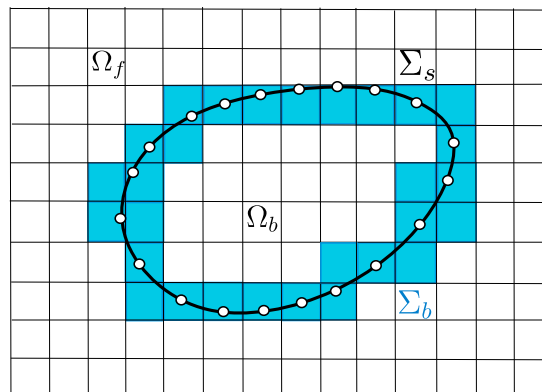


Figure 2.8: IBM regions

The Darcy's law is used in the body interface region Σ_b and the solid region Ω_b . To do so, a spatial dependent tensor $\mathbf{D}(\mathbf{x})$ must be defined. The resulting penalty term is modelled as follows:

$$\mathbf{f}_P = \begin{cases} \mathbf{0} & \text{if } \mathbf{x} \in \Omega_f \\ -\nu \mathbf{D}(\mathbf{u} - \mathbf{u}_{ib}) & \text{if } \mathbf{x} \in (\Sigma_b \cup \Omega_b) \end{cases} \quad (2.22)$$

$\mathbf{u}_{ib}(\mathbf{x}, t)$ is a target velocity representing the immersed body's physical behaviour. In our case, $\mathbf{u}_{ib} = \Omega \times \mathbf{R}$. The components D_{ij} of the tensor $\mathbf{D}$ are usually controlled to obtain the

best compromise between the accuracy and stability of the numerical solver. To converge to the correct result, D should tend to the infinite. But this lead to numerical instabilities.

We can also write Equation 2.22, once including in Naviers-Stokes equations, as :

$$(\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{u} - \underbrace{\xi \nu \mathbf{D}(\mathbf{u} - \mathbf{u}_{ib})}_{\mathbf{f}_P} \quad (2.23)$$

$$\xi = \begin{cases} 0 & \text{if } \mathbf{x} \in \Omega_f \\ 1 & \text{if } \mathbf{x} \in (\Sigma_b \cup \Omega_b) \end{cases} \quad (2.24)$$

Here, we are dividing both side of the equations by ρ for simplicity which can be considered uniform as we assume incompressible conditions. So $p = \frac{P}{\rho}$, then $[p] = M \cdot L^{-1} \cdot T^{-2}$ and $[\mathbf{u}] = L \cdot T^{-1}$

As we use a RANS turbulence model, we need also to penalize the k and ω coefficients.

Finally, the forcing terms are define as follows⁷⁸,

$$\begin{aligned} \mathbf{f}_{P_u} &= \xi \nu \mathbf{D}(\mathbf{u} - \mathbf{u}_{ib}) \\ \mathbf{f}_{P_K} &= \xi \nu D_\eta (\mathcal{K} - \mathcal{K}_{ib}) \\ \mathbf{f}_{P_\omega} &= \xi \nu D_\omega (\omega - \omega_{ib}) \end{aligned} \quad (2.25)$$

where $\omega_{vis} = \frac{6\nu}{\beta_1 y_{ref}^2}$ et $\omega_{log} = \frac{\sqrt{\mathcal{K}}}{C_\mu \kappa y_{ref}}$

The problem of this method is that it leads to a few constants that need to be tuned to obtain the best results. For that, we use the EnKF method to optimize these constants.

2.2.2 The open-source code OpenFOAM

Simulations are performed using the C++ framework OpenFOAM, used to discretize the equations previously presented. OpenFOAM is an finite-volume, open-source code which provides many tools, referred to as executables.

OpenFOAM provides a wide range of tools for solving the Navier-Stokes equations, including solvers, meshing tools, and post-processing tools. It is widely used in the CFD community and has a large user base. It's developped since 2004 and is composed today of more than a million lines.

- **Pre-processing**, including meshing and decomposing tools;
- **Solving**, including in-built and custom solvers. Each solver is built to answer a specific CFD problem;
- **Post-processing**, including many tools and libraries to compute post-processing operations. OpenFOAM fields can be visualized with **ParaView**, an open-source visualization application.

OpenFOAM is not provided with a graphical interface, which is why a simulation is built by writing “dictionaries”, in which is stored the necessary information for the simulation (mesh, turbulence model, maximum number of characteristic times, ...). See tree structure Annex 2.

⁷<https://www.simscale.com/forum/t/defining-turbulent-boundary-conditions/80895>

⁸<https://www.openfoam.com/documentation/guides/latest/doc/guide-bcs-wall-turbulence-omegaWallFunction.html>

// Ou si plus lisible : //

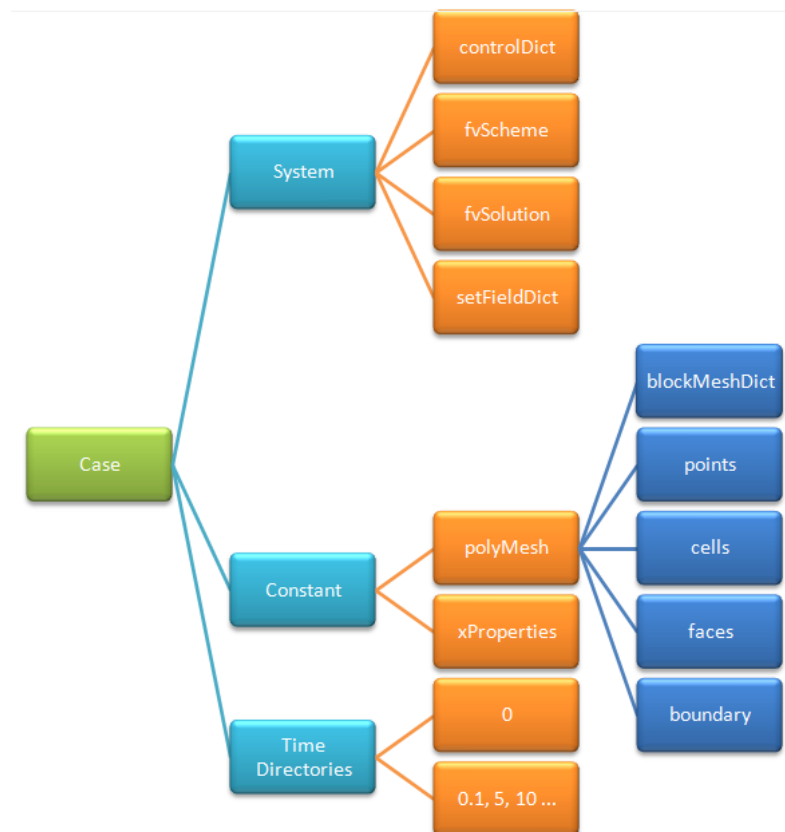


Figure 2.9: Tree of the OpenFOAM dictionaries

// Ou si plus lisible : //

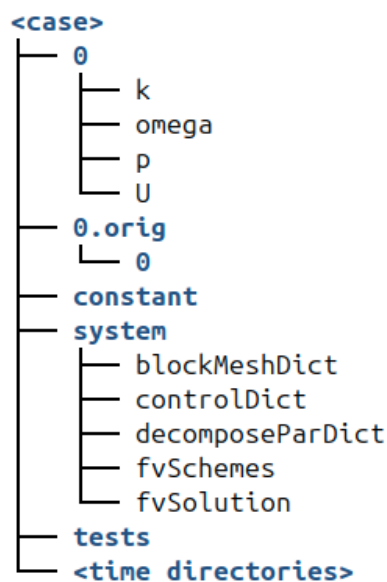


Figure 2.10: Tree of the OpenFOAM dictionaries

The **system** directory contains the main parameters concerning the progress of the simulation:

- The **controlDict** dictionary defines parameters such as start and end time, time steps, data output formatting, or data average computation;

- the **blockMeshDict** dictionary defines the computational domain, mesh and boundary conditions implementation;
- the **decomposeParDict** dictionary defines the geometry and mesh decomposition, with the objective to run the simulation in parallel, using multiple cores;
- the **fvSchemes** dictionary defines the discretisation schemes used in the CPUs equations;
- the **fvSolution** dictionary defines the equation solvers, tolerances and other algorithm controls.

The **constant** directory contains the different physical properties of the simulation. Moreover, the subfolder **polyMesh** contains the mesh data, obtained after the **blockMesh** command line is run:

- The **momentumTransport** dictionary defines the simulation type (DNS, RANS or LES) and, in the case of RANS or LES, specifies the corresponding turbulence closure or subgrid-scale model.
- the **transportProperties** dictionary defines the transport model (e.g., Newtonian) and some transport coefficients, such as the kinematic viscosity ν ;
- the **MRFProperties** dictionary defines in our case to specify the rotating zone with the origin, rotation axis and rotation speed.

The **0** directory contains the initial solution's fields and the Boundary Conditions (BC) as well. A compiled solver can be called in the terminal of the case directory to run the simulation. If it's called, the **postProcessing** directory stores data from the running simulation, such as forces and probes' data. The **processor0...N-1** directories directly result from the geometry decomposition, which is proceeded through the **decomposeParDict** dictionary. The **dummy.foam** is an empty file which allows to visualize the simulation with ParaView.

More precisely, for example, to implemente the IBM forage term on the speed "u", we can manually modify the momentum equation of the corresponding solver as follows:

```

1 // Momentum predictor step
2
3 MRF.correctBoundaryVelocity(U) // Corrects U at boundaries to include MRF effects
4                               // (e.g., rotating walls)
5
6 tmp<fvVectorMatrix> tUEqn
7 (
8     fvm::div(phi, U)           // Convective term
9     + MRF.DDt(U)               // Time derivative in the rotating reference frame
10    + turbulence->ddivDevSigma(U) // Turbulent diffusion term
11    + cmptMultiply(Dprime, U - UIB) // Forcing term: Dprime * (U - UIB)
12    ==
13    fvModels.source(U)         // Source terms from external models
14                               // (e.g., body forces), here equal to 0
15 );
16 fvVectorMatrix& UEqn = tUEqn.ref();
17
18 UEqn.relax() // Applies under-relaxation for numerical stability
19
20 fvConstraints.constrain(UEqn); // Applies physical constraints
21                               // (e.g., boundary conditions)

```

With

- *cmptMultiply* the element-wise matrix multiplication;

- $D_{prime} = \xi \nu D$, with D_{prime} denoting the reduced form of D . The prime symbol is used to distinguish the non-dimensional quantity from its physical counterpart;
- U the flow velocity;
- UIB is the imposed velocity at the walls.

2.2.3 EnKF

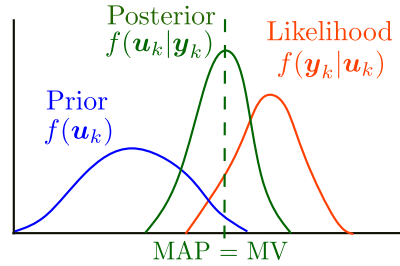


Figure 2.11: Kalman Filter algorithm scheme

As seen in the section 2.1, we are going to use the stochastic EnKF to assimilate the high-fidelity data. But first, let's explain the Kalman Filter⁹.

Kalman Filter The estimation is calculated by using a set of observations $\mathbf{y}$, a prior state $\mathbf{x}$, and a linear model $\mathbf{M}$. The estimation state is obtained through two operations, the forecast and the analysis, respectively using the superscript f and a .

The system state $\mathbf{x}_k$ is as follows:

$$\begin{bmatrix} \mathbf{u}_k \\ \boldsymbol{\theta}_k \end{bmatrix} \in \mathbb{R}^{(n_{\text{vars}} \cdot n_{\text{cells}} + n_{\boldsymbol{\theta}}) \times m} \quad (2.26)$$

where:

- $\mathbf{u}_k$ is the state matrix (can contain velocities, pressure, temperature...) of the domain at the time instant k , which we don't care about in our case;
- $\boldsymbol{\theta}_k$ are the model coefficients, assumed time-invariant in our case;
- m is the number of ensemble members.

The observation data $\mathbf{y}_k$ is defined as:

$$[\mathbf{y}_k]_{n_{\text{obs}}, m} \sim \mathcal{N}(y_k, \mathbf{R}_k) \quad (2.27)$$

Observation data may be built using:

- Velocity, pressure, temperature and more... values;
- Some discret model parameters such instantaneous global force coefficients C_D , C_L , C_f . In our case, It will be five IBM forcing parameters.

⁹https://cones-dev-pe431-2dbff22738a4d54c2b947e6fa9a9a77fadd187b3bd1a85c.gitlab-pages.ensam.eu/data_assimilation.html

Prediction step:

$$\hat{\mathbf{x}}_{k|k-1} = \mathbf{M}_k \hat{\mathbf{x}}_{k-1|k-1} + \mathbf{B}_k \mathbf{u}_k \quad (2.28)$$

$$\mathbf{P}_{k|k-1} = \mathbf{M}_k \mathbf{P}_{k-1|k-1} \mathbf{F}_k^\top + \mathbf{Q}_k \quad (2.29)$$

Update step:

$$\tilde{\mathbf{y}}_k = \mathbf{z}_k - \mathbf{H}_k \hat{\mathbf{x}}_{k|k-1} \quad (2.30)$$

$$\mathbf{S}_k = \mathbf{H}_k \mathbf{P}_{k|k-1} \mathbf{H}_k^\top + \mathbf{R}_k \quad (2.31)$$

$$\mathbf{K}_k = \mathbf{P}_{k|k-1} \mathbf{H}_k^\top \mathbf{S}_k^{-1} \quad (2.32)$$

$$\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k \tilde{\mathbf{y}}_k \quad (2.33)$$

$$\mathbf{P}_{k|k} = (\mathbf{I} - \mathbf{K}_k \mathbf{H}_k) \mathbf{P}_{k|k-1} \quad (2.34)$$

Where:

- $\hat{\mathbf{x}}_{k|k-1}$: predicted state at k
- $\hat{\mathbf{x}}_{k-1|k-1}$: updated state at $k-1$
- $\mathbf{M}_k$: state transition matrix
- $\mathbf{B}_k$: control input matrix
- $\mathbf{u}_k$: control input
- $\mathbf{P}_{k|k-1}$: predicted covariance
- $\mathbf{P}_{k-1|k-1}$: updated covariance at $k-1$
- $\mathbf{Q}_k$: process noise
- $\mathbf{z}_k$: measurement
- $\mathbf{H}_k$: observation matrix
- $\tilde{\mathbf{y}}_k$: innovation
- $\mathbf{S}_k$: innovation covariance
- $\mathbf{K}_k$: Kalman gain
- $\mathbf{R}_k$: measurement noise
- $\mathbf{I}$: identity matrix
- $\hat{\mathbf{x}}_{k|k}$: updated state
- $\mathbf{P}_{k|k}$: updated covariance

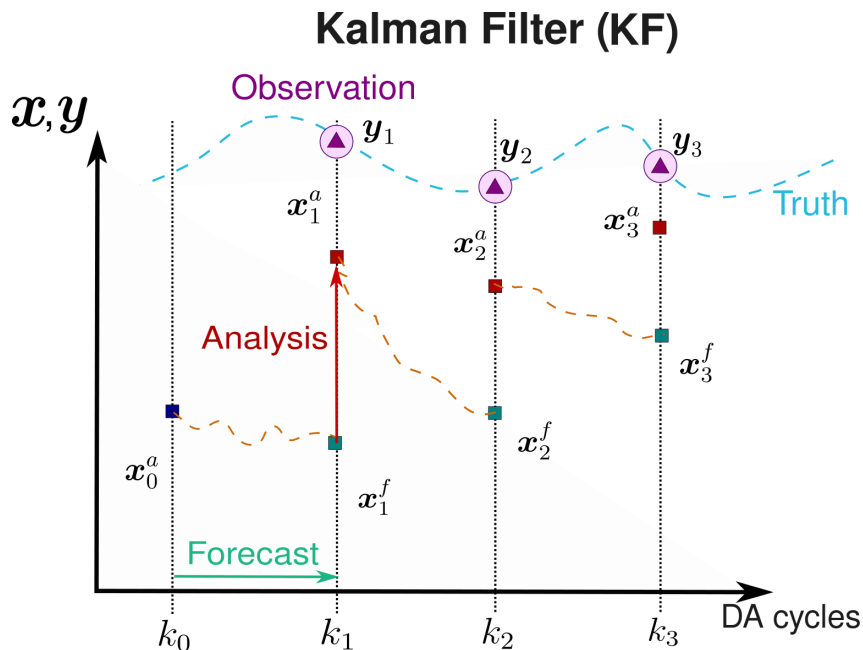


Figure 2.12: Kalman Filter algorithm scheme

Kalman filter present two main drawbacks that concern our case. First, as the Extended Kalman Filter (EKF), it's only works for moderate deviations from linearity and Gaussianity. If M represent the Navier-Stokes equations, then is very not linear. Secondly, for a CFD mesh, the covariance matrix $\mathbf{P}_k$ would have a dimention of $(n_{vars} \times n_{cells}, n_{vars} \times n_{cells})$.

The EnKF allows high dimensional systems by avoid the explicit definition of $\mathbf{P}_k$ and it's Monte-Carlo realisations are much more design for non-linear models. A problem can be the underestimation of $\mathbf{P}_k^a$ but the inflation, define later on this subsection, and localisation, helps mitigate this issue.

Ensemble Kalman Filter We saw previously in this subsection that today, we can theoretically run a DNS which resolves exactly the Navier-Stokes equations, but it is still too expensive to be used in practice. Therefore, we generally use LES or RANS models to simulate the flow. The EnKF method can be applied to these models to adjust specific model parameters or flow fields, enhancing the overall physical consistency of the simulation.

A good example based on the simple and academic *cavity test case* illustrates it well. The cavity test case is a simple flow in a square cavity, where the flow is driven by a moving lid. The flow is characterized by a recirculation zone and a vortex in the center of the cavity. The EnKF method can be used to assimilate data from the flow, such as velocity or pressure measurements, to improve the accuracy of the simulation. The EnKF method can also be used to estimate the parameters of the model, such as viscosity or turbulence model coefficients, to improve the physical accuracy of the simulation.

In our test case, we are going to optimise the velocity wall condition. To do so, we use the result on 5 locations of a precise converged simulation with a to boundary condition of 5 m.s^{-1} as a reference. We will then assimilate data from the flow, at the 5 observation points (see Figure 2.13a). We start from a simulation with a wall condition of 1 m.s^{-1} , which is not the correct value. The EnKF method will then be used to correct the wall condition by assimilating the data from the flow at the 5 observation points. The EnKF method will iteratively update the wall condition until it converges to the correct value, which is 5 m.s^{-1} in this case. In only 3 EnKF iterations, the wall condition converges to the correct value (with a relative error of 0.05 m.s^{-1}).

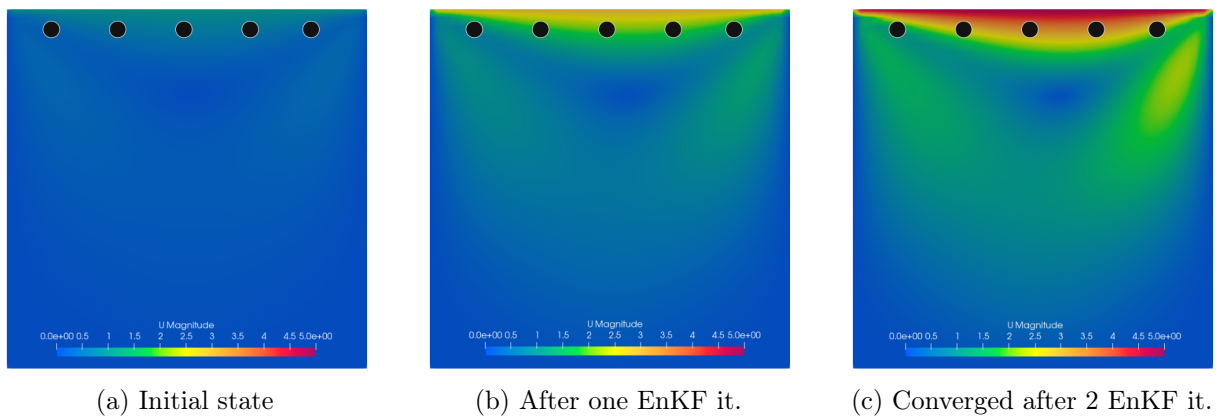


Figure 2.13: Evolution of the state through EnKF iterations (●: observation point)

The following matrices are needed to build the EnKF algorithm:

1. **Observation covariance matrix $\mathbf{R}_k$** : we assume Gaussian, non-correlated uncertainty for the observations ($\sigma_{i,k}$ is the standard deviation for the variable i at time instant k).

$$[\mathbf{R}_k]_{n_{\text{obs}}, n_{\text{obs}}} = \sigma_{i,k}^2 [\mathbf{I}]_{n_{\text{obs}}, n_{\text{obs}}}$$

2. **Sampling matrix** $\mathcal{H}(\mathbf{x}_i^f)$: it is the projection of the model into the position of the observations.

$$\left[\mathcal{H}(\mathbf{x}_i^f) \right]_{n_{\text{obs}}, m}$$

3. **Anomaly matrices** for the system's state $\mathbf{x}_i^f$ and sampling $\mathcal{H}(\mathbf{x}_i^f) = \mathbf{s}_i^f$ matrices. They measure the deviation of each realization with respect to the mean.

$$\mathbf{X}_i = \frac{\mathbf{x}_i - \bar{\mathbf{x}}}{\sqrt{m-1}}, \quad \mathbf{S}_i = \frac{\mathbf{s}_i - \bar{\mathbf{s}}}{\sqrt{m-1}}$$

with

$$[\mathbf{X}]_{n_{\text{vars}} \times (n_{\text{cells}} + n_{\theta}), m}, \quad [\mathbf{S}]_{n_{\text{obs}}, m}$$

4. **Kalman Gain matrix**:

$$\mathbf{K} = \mathbf{X} \mathbf{S}^T (\mathbf{S} \mathbf{S}^T + \mathbf{R})^{-1}$$

with

$$[\mathbf{K}_k]_{n_{\text{vars}} \times (n_{\text{cells}} + n_{\theta}), n_{\text{obs}}}$$

For RANS representation, there is no physical time advancement of the model, so each instant is represented through a different iteration of the solver, and

Algorithm 1 Scheme of the EnKF used in the present analyses for parameter optimization.

- 1: **Input:** $\mathcal{M}$ (model); $\mathbf{R}$ (observation covariance); $\mathbf{y}_k$ (observations);
 $\mathbf{u}'_{i,0} = [\mathbf{u}_{i,0}, \theta_{i,0}]^T$ (initial prior states with $\mathbf{u}_{i,0} \sim \mathcal{N}(\bar{\mathbf{u}}_0, \mathbf{P}_0)$ and $\theta_{i,0} \sim \mathcal{N}(\bar{\theta}_0, \mathbf{P}_0)$)
 $k = 1, 2, \dots, K$ is the iteration index; $i = 1, 2, \dots, N_e$ is the ensemble index
 - 2: **for** $k = 1$ to K **do**
 - 3: **for** $i = 1$ to N_e **do**
 - 4: Advance in time: $\mathbf{u}_{i,k}^f = \mathcal{M}(\theta_{i,k-1}) \mathbf{u}_{i,k-1}$
 - 5: Observation: $\mathbf{Y}_{i,k} = \mathbf{y}_k + \mathbf{e}_i$, $\mathbf{e}_i \sim \mathcal{N}(0, \mathbf{R})$
 - 6: Compute ensemble means: $\bar{\mathbf{u}}_k^f = \frac{1}{N_e} \sum_{i=1}^{N_e} \mathbf{u}_{i,k}^f$, $\overline{\mathcal{H}(\mathbf{u}_k^f)} = \frac{1}{N_e} \sum_{i=1}^{N_e} \mathcal{H}(\mathbf{u}_{i,k}^f)$
 - 7: Anomaly matrices: $\mathbf{X}_k = \frac{\mathbf{u}_{i,k}^f - \bar{\mathbf{u}}_k^f}{\sqrt{N_e-1}}$, $\mathbf{S}_k = \frac{\mathcal{H}(\mathbf{u}_{i,k}^f) - \overline{\mathcal{H}(\mathbf{u}_k^f)}}{\sqrt{N_e-1}}$
 - 8: Kalman gain: $\mathbf{K}_k = \mathbf{L} \odot \mathbf{X}_k (\mathbf{S}_k)^T [\mathbf{S}_k (\mathbf{S}_k)^T + \mathbf{R}]^{-1}$ /// L to change ///
 - 9: Update: $\mathbf{u}_{i,k}^a = \mathbf{u}_{i,k}^f + \mathbf{K}_k (\mathbf{Y}_{i,k} - \mathcal{H}(\mathbf{u}_{i,k}^f))$
 - 10: (Deterministic) covariance inflation: $\mathbf{u}_{i,k}^a = \bar{\mathbf{u}}_k^a + \lambda_k (\mathbf{u}_{i,k}^a - \bar{\mathbf{u}}_k^a)$
 - 11: **end for**
 - 12: **end for**
-

Observation data y can be some local sensors (such as PIV data) or global variables (such as the performance)

$$\text{With } \bar{\mathbf{u}}_k^f = [\bar{\mathbf{u}}_k^f, \bar{\theta}_{k-1}]^T \text{ and } \mathbf{u}_{i,k}^f = [\mathbf{u}_{i,k}^f, \theta_{i,k-1}]^T$$

Choisir entre "u" et "x"

$$\text{In our case, } \bar{\mathbf{u}}_k^f = [\bar{\theta}_{k-1}]^T, \mathbf{u}_{i,k}^f = [\theta_{i,k-1}]^T \text{ and } \lambda_k = 0, \mathbf{y}_k = \mathbf{y}.$$

Some *deterministic inflation* is included to increase the variability of the state predicted by the EnKF:

$$\mathbf{u}_{i,k}^a = \bar{\mathbf{u}}_k^a + \lambda_k (\mathbf{u}_{i,k}^a - \bar{\mathbf{u}}_k^a) \quad (2.35)$$

//// Illustrer l'inflation déterministic : ////

2.2.4 The CONES library

CONES is a library coded in Python/C++, stands for "Coupling OpenFOAM with Numerical EnvironmentS" and has been developed since 2022 at LMFL. Is the only code that permits online DA in CFD codes (it uses Message Passing Interface (MPI) process to do it). It's parallelized and can be used on supercomputers. It is mainly designed to work with OpenFOAM and DA codes as EnKF.

A specific directory structure must be observed, as outlined in Figure 2.14. Let's take our OpenFOAM/EnKF example as support to explain CONES. Concretely, on one side, m OpenFOAM simulations are running with run at the same time on n processors and on the other side, the EnKF run on e processors (usually only one is enough). At the beginin, all variables are initialized on both sides, using MPI process. Then, all simulation run at same time (with different initial states). Sampled data $(H(u_k^f))$ and $\begin{bmatrix} \mathbf{u}_k \\ \boldsymbol{\theta}_k \end{bmatrix}^f$ are send to the EnKF. Using y_k , EnKF algorithm can computes and send back the updated velocity field (not computed in our case) and the parameters θ_k^a to the different ensemble members (each are different before converging). With thi new parameters, the loop can be done again.

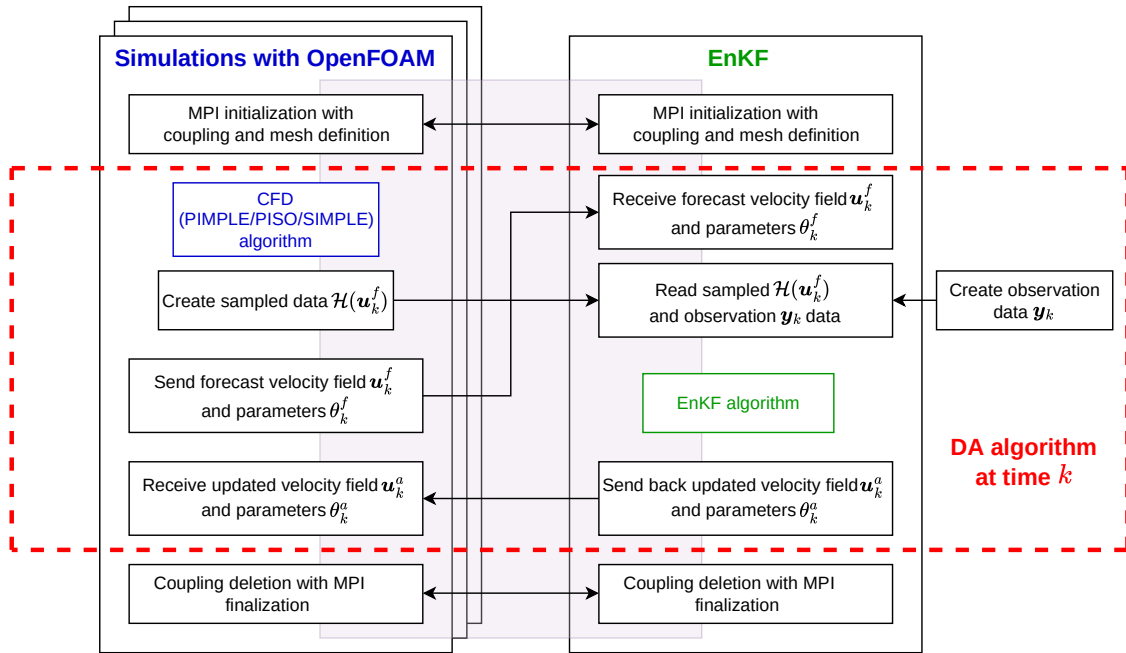


Figure 2.14: CONES structure

2.2.5 Pump modelling

Figure 2.15 illustrates the model of the pump on a yz plane, for $x = 0$. The suction pipe has a length of $H = 10R_1$, allowing the near-wall boundary layer to develop. Then, a cylinder containing the impeller is defined, where the IBM and MRF will be applied. At the entrance of the impeller - outlet of the suction pipe junction and at the outlet of the impeller - inlet of the diffuser junction, there was leakage as the whole impeller was moving during the experiment. We assume a zero-leakage as we aren't in a very high mass flow rate.

In our simulation, the z -axis aligns with the rotational axis, and the reference frame is centred at the impeller. The plane at $z = 0$ divides the diffuser into two identical halves.

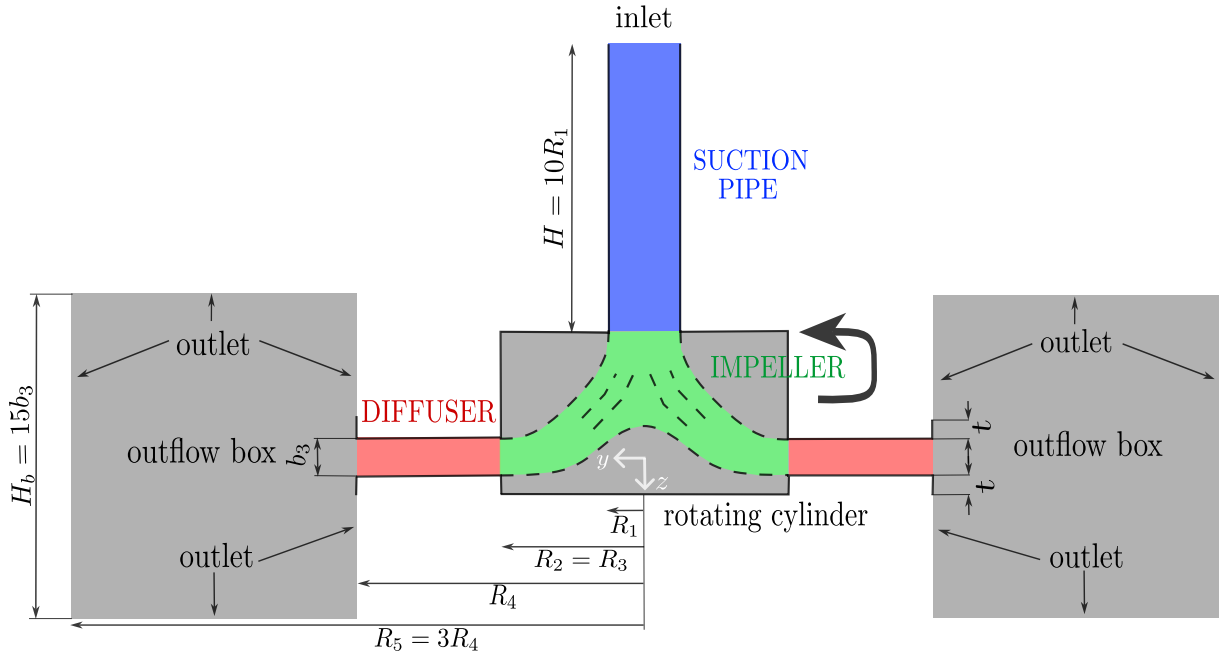


Figure 2.15: Model of the RESEDA benchmark

Boundary conditions are defines as that:

- Inlet: velocity inlet
- Outlet: pressure outlet
- Wall: no-slip wall

The penalisation coefficients used in the Equation 2.25 are defines as follow :

$$\begin{aligned} \mathbf{u}_{ib} &= \mathbf{u}_r + \boldsymbol{\Omega} \times \mathbf{r} \\ \mathcal{K}_{ib} &= 0 \\ \omega_{ib} &= (\omega_{vis}^2 + \omega_{log}^2)^{1/2} \end{aligned} \quad (2.36)$$

To process CFD, we need to discretize the numerical model of the pump. The mesh is generated using the **blockMesh** utility, which is a built-in OpenFOAM tool. It is composed of around 11 million hexahedral cells. (quasi only, exepted some interface, cf Figure 2.16). We don't need complex mesh because we dont neek to fit the geometry of the impeller as we use IBM.

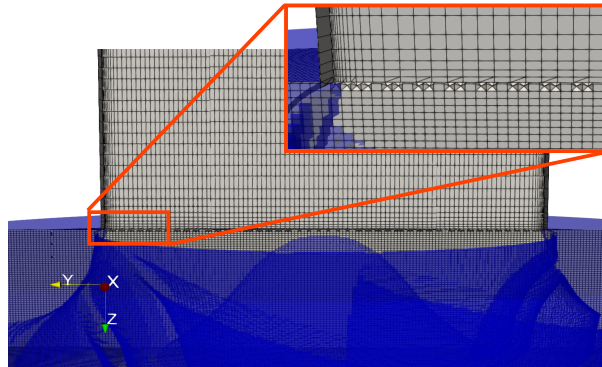


Figure 2.16: Mesh interface between the pipe and the impeller

// Parler de y^+ //

2.3 Experimental data and pre-processing

With the conditions specified in the /// là où il y a le tableau en question ///, and a value of $3e7$ for the all five forcing parameters, we obtain these velocity and pressure profiles.

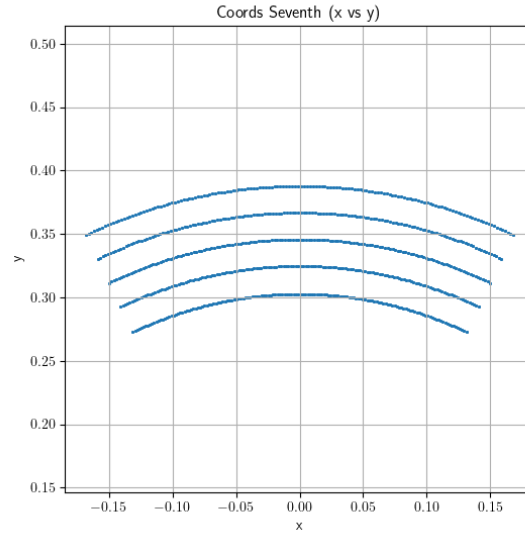


Figure 2.17: Interpolation points over 360/7 degrees

As seen in /// Figure a référer ou mettre ici ///, Figure 2.22, we use five observations points on the three PIV plans. We are able to plot this curves, comparing the velocities of the simulation and of the PIV data.

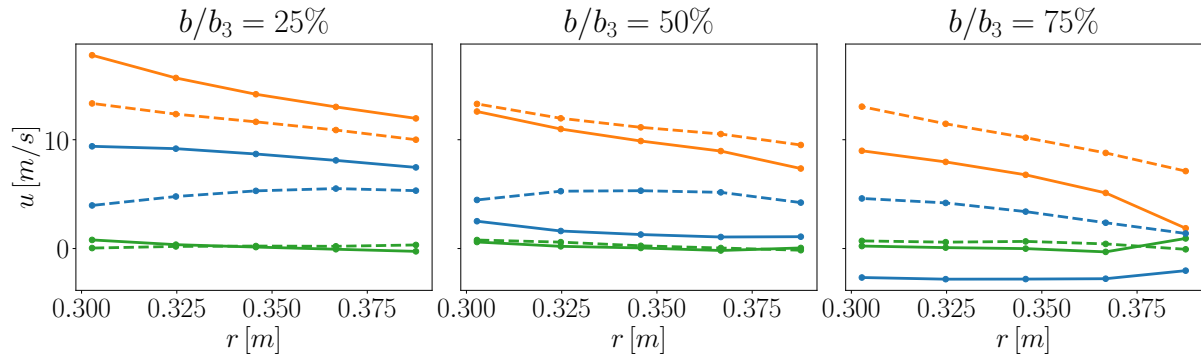


Figure 2.18: Comparison between the profiles inside the diffuser for the numerical simulation (continuous lines), and the PIV data (dashed lines). In particular, (—) represents the radial velocity u_r , (—) depicts the azimuthal velocity u_θ , and (—) illustrates the vertical velocity u_z .

Our goal is to make these curves fit as well as possible. We already see that there is a notable difference between the radial velocity in the plan at 75%. The numerical simulation predict a negative velocity instead of a positive one. This is explained by a non physical recirculation bubble as you can see in the

A good indicator of the accuracy of the simulation is the performance of the pump. It's directly linked to the pressure difference between the outlet and the inlet. As seen // previously in this chapter ?? //, we should obtain around 415Pa. In our case, we obtain a pressure difference of around 306Pa, that is far from what we expect.

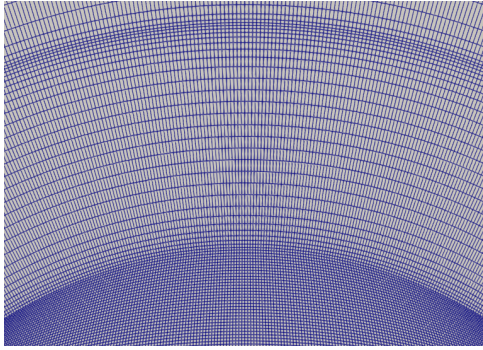
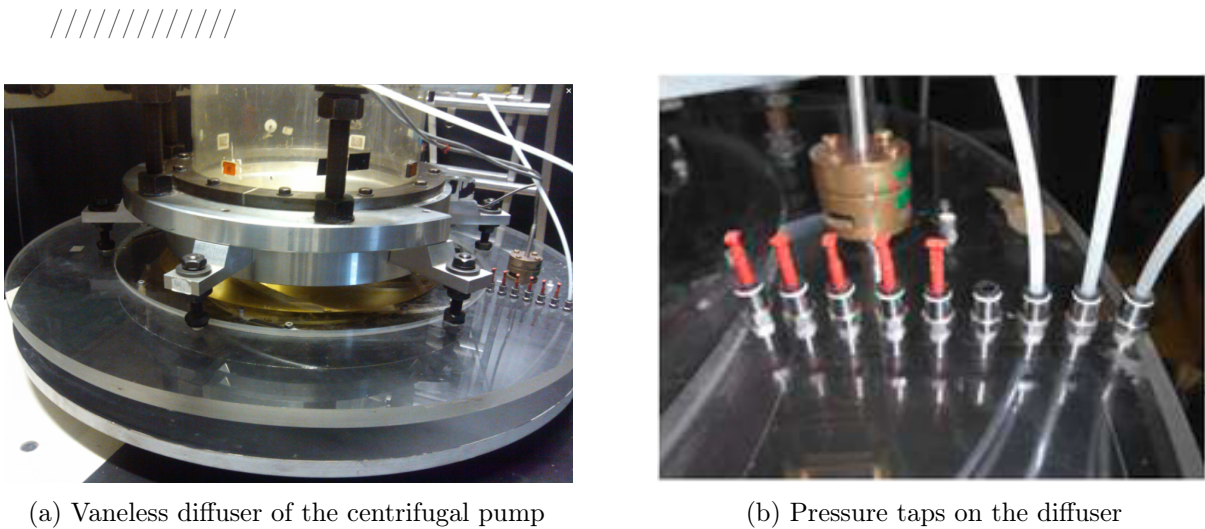


Figure 2.19: Diffuser Mesh



(a) Vaneless diffuser of the centrifugal pump

(b) Pressure taps on the diffuser

Figure 2.20: Diffuser and pressure taps of the centrifugal pump

The operating conditions of the machine were :

Impeller characteristics		
R_1	Tip inlet radius	141.1 mm
R_2	Outlet radius	257.5 mm
Z	Number of blades	7
Q_d	Design flow rate	$0.236 \text{ m}^3/\text{s}$
K	Mean blade thickness	9 mm
Ω	Rotating velocity	1 200 rpm
β_2	Outside blade angle (from peripheral velocity $U = R_2\Omega$)	22.5°
Vaneless diffuser characteristics		
b_3	Diffuser width	38.5 mm
R_3	Inlet radius (without leakage, i.e., $R_3 = R_2$)	257.5 mm
R_4	Diffuser outlet radius	385.5 mm
t	Thickness of the upper and lower sections of the diffuser	20 mm

Table 2.1: Summary of the main geometrical and operating parameters of the impeller and diffuser at design conditions

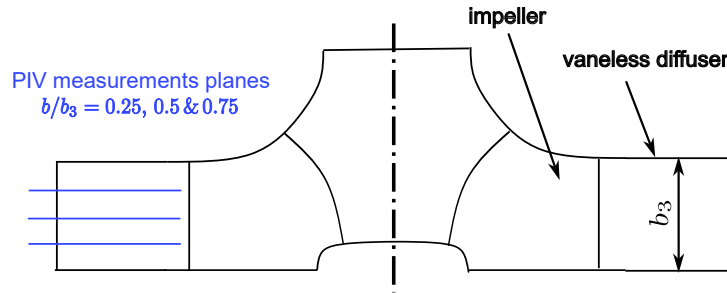


Figure 2.21: Position of the PIV planes on the diffuser

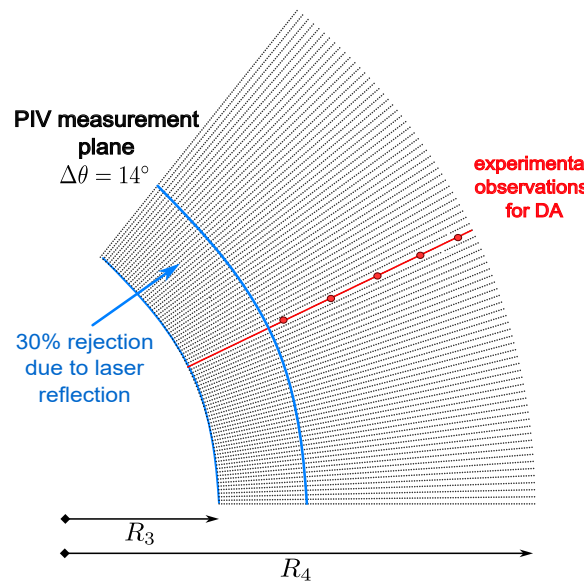


Figure 2.22: Position of the PIV points on a polar slice of the diffuser

////////////////

PIV data Data is stored in a directory structure that contains six directories. The six directories correspond to two tests realised at 25%, 50% et 75% of the diffuser height. Each test was repeated twice for the same operating condition and the same height (R1, R2). In each directory, there are 1581 files corresponding to instantaneous ASCII data file. The ten columns are corresponding to

$$x(m), y(m), V_x(m.s^{-1}), V_y(m.s^{-1}), r(m), \theta(rad), Vr(m.s^{-1}), V_\theta(m.s^{-1}), V_z(m.s^{-1}), Unknown$$

And the 10,125 lines correspond to the 81 locations over x and 125 locations over y of the PIV. Processing of these data is detailed in the // This section ?? //

Pressure data In addition to the nine pressure tapes on the diffuser, M. Fan has also captured (Figure 2.23) the inlet and outlet pressure that allows to compute the global performance of the pump.

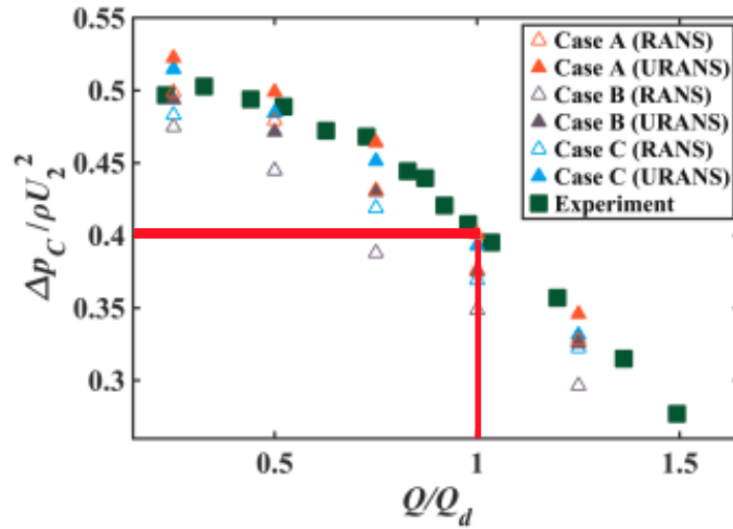


Figure 2.23: Overall pump performance versus reduced pump flow rate Q

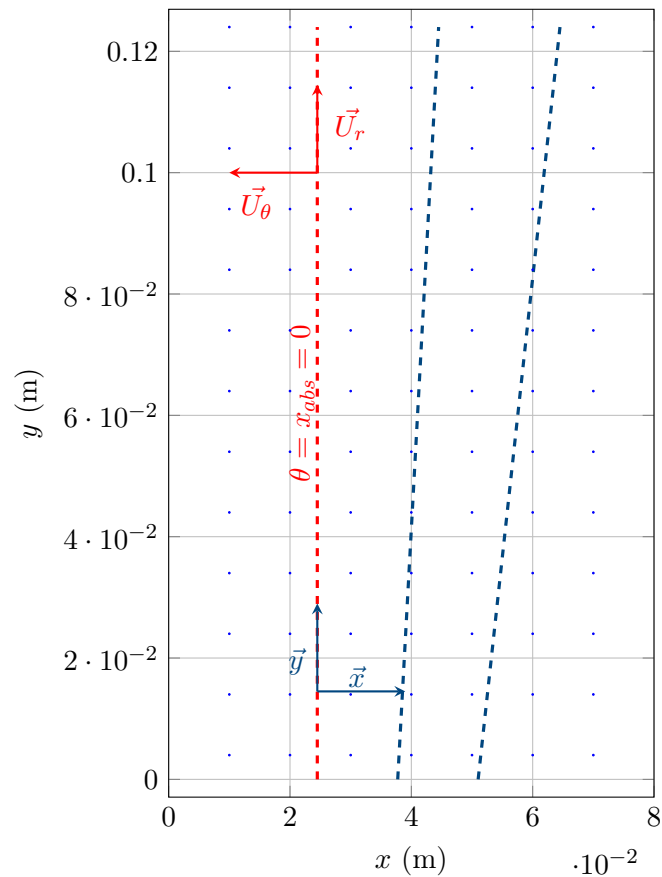
As we are at $Q/Q_d = 1$, we can retrieve

$$\frac{\Delta P}{\rho U^2} = 0.4 \Rightarrow \Delta p = 0.4 \times \rho \times U^2 = 0.4 \times 1.225 \times 32.24^2 \approx 415 \text{ Pa}$$

//////////

The PIV data is acquired in a rectangular zone of 0.08 m x 0.124 m. The cathesian coordinates are given in a local system, where the absolute origin is at the center of the pump.

Zone d'acquisition PIV



Dans la simulation numérique, l'axe z est vers le centre de la Terre, contrairement à l'expérience où il est vers le haut. La roue tourne dans le sens trigonométrique dans le repère de la simulation numérique et dans le sens antitrigonométrique dans l'expérimentation. Actuellement, je respecte le formalisme de la simu num pour le repère.

Parler des coefficients de relaxation.

1. First, we need to know how to convert the coordinates from the local system to the absolute system. We find that between the line 55 and 56, the radius is constant and pass from lower than π to upper. In fact, for the plan at 75%, it is the same approach but between the line 65 and 66 because of some technical experimental constraints. We also did plot x over theta for three different values of theta and find out an intersection point for $x = 0.0245$. So now, we know that for $x = 0.0245$, $\theta = \frac{\pi}{2}$

$$\begin{cases} x_{rel} = x_{abs} + 0.0245 \\ y_{rel} = x_{abs} + R_2 \end{cases} \quad (2.37)$$

2.Average For now we are in stationary case (time invariant). So we are going to deal with data average over time. There are around 49 maps (=file) by turn. So to average the data over the time, we can average over the ϕ (ϕ appartient à 0-211rd) (=maps). That into the six directory.

3.Sampling We will also average data between R1 and R2 to go to 3 files (of 10x81x125 elements). Now, we keep only the data on the line $x = 0.0245$ (in fact we average data between $x = 0.024$ and $x = 0.025$). Data are then truncate at line 89, where the radius is upper than 0.3, because of some laser reflexion on the pump during the PIV experiment.

4.Shaping for CONES Cones keep data in a particular way. We need to convert the data in a format that cones can read. From 3 files (of 10x81x125 elements), to the files `fieldCylindrical.txt` and `obs_coordinates.txt`. Describe shapes...

5.Creation of a file for the interpolation Finally, we need to create an `coords_seventh.txt` file for the interpolation function in CONES. We don't just use the coordinates of the points in the PIV data, because we want to average over the angle θ on the pump. The numerical simulation is $\frac{2\pi}{7}$ OU $2\pi/7$ periodic, so we interpolate in 150 points over theta between 0 and $2\pi/7$. To do that, we just take the `obs_coordinates.txt` file and duplicate 150 times the points over the angle θ (between $-\pi/7$ and $\pi/7$). In our case, we chose 150 that will approximately correspond to the shape of a cell at this radius. To be consistent with axis further, we take points between $-\pi/7 + \pi/2$ and $\pi/7 + \pi/2$

Here is the plot of the experimental data and the numerical data (for $D = 3e7$ (introduire la variable)), at the same points (both are averages over time and over the angle θ).

////////

Limitations IBM à améliorer avec DA Grâce aux données xp (mentionné partie 2.1.2, on peut faire de la DA)

2.4 IBM preliminary results

Explication de coords_seventh.txt, coords_360.txt et coords_7.txt

In this work, we aim to improve the numerical model of a centrifugal pump using experimental data obtained from PIV. On one side, we have a numerical simulation of the pump (the digital twin), and on the other, experimental PIV measurements. To combine the strengths of both, we use data assimilation, a method that updates a numerical model using real-world data. We specifically use the EnKF, a data assimilation technique that relies on an ensemble of simulations to estimate and reduce uncertainty in the model. The process can be summarized in two main steps:

1. **Forecast Step** We run a steady-state numerical simulation of the pump using the Multiple Reference Frame (MRF) and the IBM methods. This yields an ensemble of model states, representing the expected flow field within the pump.
2. **Analysis Step** We compare the simulation results with the experimental PIV data. To do this, we interpolate the PIV measurements onto the mesh of the numerical model, creating an ensemble of observation vectors. We then compute the difference between the simulated and measured data, known as the innovation (see mathematical formulation in subsection 2.2.1). This innovation is used to update the ensemble through the EnKF algorithm, improving the numerical model.

The measurements are done as follow :

1. **Nature of the Numerical Simulation** The simulation is stationary (time-independent) and uses the MRF approach to model rotation. While the velocity field remains constant over time, it varies in space—especially in the diffuser region. Because of the pump's geometry, we expect a periodic velocity pattern that repeats every 1/7th of a revolution (due to the 7 blades). However, this repeating pattern is not clearly visible in unsteady simulations due to wake deformation.
2. **Nature of the Experimental Data** The PIV system captures 49 images per full rotor revolution, meaning there are 7 images between two successive blades. In the rotating frame of reference, each image corresponds to a phase shift of approximately $360^\circ/49$.

Ideally, to compare consistent datasets, we could run a time-resolved unsteady simulation and synchronize it with the PIV snapshots. However, for this initial study, we chose to work with a steady MRF simulation.

Although we could attempt to reconstruct the spatial velocity field from the phase-averaged PIV data (as done in [citer article Antoine]), there are two main limitations:

To address these issues, we opted to average the PIV data over time, which is equivalent to spatial (azimuthal) averaging in the rotating frame. On the simulation side, we therefore perform azimuthal averaging of the numerical results to make them consistent with the experimental data.

This approach ensures that the data fed into the EnKF—both the forecast and the observations—are comparable and compatible in both spatial and statistical sense.

potentiellement en annexe

Concretely, to average the data over the angle θ , we need to modify the interpolation function in CONES. Instead of read coords, the interpolation function read the data in the coords_seventh.txt file.

Then, it computes the interpolation and average the data over the angle θ (between $-\pi/7$ and $\pi/7$), by doing a somme over θ and dividing by the number of points (150). To go back from $150 \times 5 \times 3 \times 3$ (to detail) to $5 \times 3 \times 3$ data. All of those computation are mad in a cylindrical referencial. Ref au schéma tikz

A sixth coefficients DP has been add. It's represent the difference of pressure between the inlet of the pipe and the outlet of the diffuser. The numerical coefficient are interpolating, respectively at $(0,0,-0.35)$ and $(0,0.390,0)$. For the high-fidelity value, we can find experimental data form the same pump [M. Fan 2022], that's match with M. Fan simulation (eq to approximately $\eta = 0.4$). So can also compute :

U_2 is the impeller tip speed computed from the impeller diameter and the rotation speed of the pump.

$$\frac{\Delta P_P}{\rho U_2^2} = \eta \Leftrightarrow \Delta P_P = \rho U_2^2 \eta \quad (2.38)$$

$$U_{2carré} = (\omega * R_2)^2 = (2 * \pi * 1200 / 60 * 0.2566)^2 = 32,2 \text{ m/s}$$

As we are in a steady simulation, the restult of the numerical pressure is divided by the density of the fluid.

$$\Delta P_P = 32.2^2 \times 0.4 = 415 Pa \quad (2.39)$$

So the state vector is now 46. Incertitude on p will be less to take more take it account. Inflation...

At pump flow rate = pump design flow rate So $Q^*/Q^* = 1$

2.4.1 Modifications on OpenFOAM

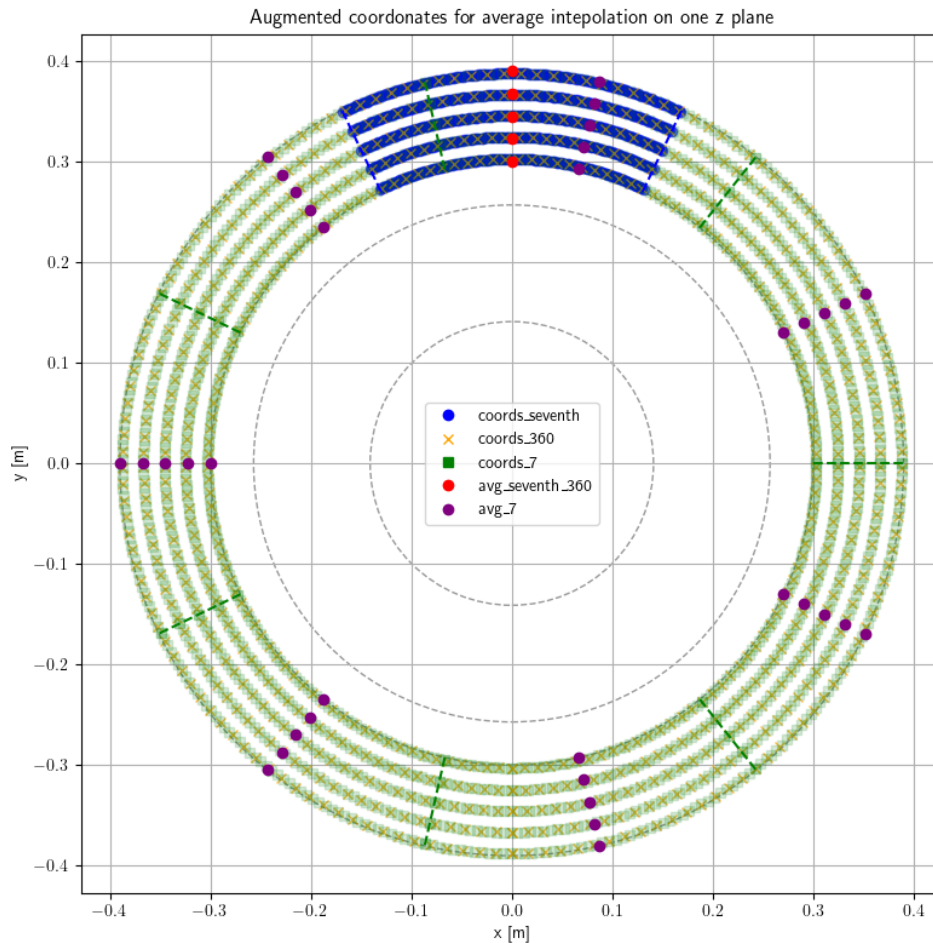


Figure 2.24

Imager avec des illustrations et vue en coupe ParaView

Pour cela j'ai commencé en moyennant 1/7 mais les paramètres étant locaux (non globaux), j'ai ensuite moyenné sur 360 degrés, ce qui permet de mieux prendre en compte les variations locales des paramètres. Cependant, nous pouvons converger vers une moyenne physique qui atteints un minimum local mais qui ne respecte pas d'avoir 7 fois le même pattern. Cela permet à F_x d'être très différent de F_y mais aussi à la simulation de diverger avec des paramètres et vitesses erronées. Pour palier ce problème, je compare la moyenne des patternes sur mes 7 segments de 360/7 degrés.

Nous avons fait l'EnKF sans l'état... Sans inflation pour commencer. Si on converge vers une solution qui ne nous satisfait pas, on peut essayer d'ajouter... Cela permet de s'échapper d'un minimum local en ajoutant du bruit artificiellement.

Nous avons fait une simu RANS mais nous aurion pu faire une simu URANS (ou LES ou DDES) avec un modèle de turbulence...

L'expérience montre un ordre de grandeur de centaines d'itération entre les phases d'analyse, mais cela dépend beaucoup du cas. Il en est de même pour le nombre de phases d'analyse. Supposons 1000 itération en moyenne entre chaque phase d'analyse, et 300 phases d'analyse, cela nous fait 300 000 itérations à simuler, soit 93h, sur env. 2800 coeurs, soit 264kh.processeur de calcul.

Comment calculer la variance que je veux mettre ? Expérience : 5% De mon coté : je vise

qu'en moyenne un membre d'ensemble ait un écart de $1e7$ par rapport au temps précédent (pour des valeurs autour de $5e7$), donc, pour 30 membres d'ensemble, environ $95\% = 2 \text{ sigma}$ dans $5e7 \pm 1e7$, c'est-à-dire $\sigma = 0.5e7 = 10\%$. Ensuite, le nombre d'itération entre les phases d'analyse est déterminé en regardant combien d'itération sont nécessaires pour faire varier la solution dans le diffuseur (influe, rapidement, converge doucement mais ce qui nous intéresse c'est que ça prenne la tendance) lorsqu'un paramètre de forçage est modifié (supposé très inférieure pour la turbulence car appliquée de partout)

The observation window was defined by observing the number of iterations required for the influence of the IBM to reach the diffuseur and take the form of the converged result. We are not interested in the turbulence parameters to determine the observation window because they are present throughout the domain, their influence is almost direct (not necessarily their convergence). We chose 800 iterations

Pb de la DA : plus rapide que DNS mais toujours plus lent que RANS et le temps d'attente et de calcul sur supercalculateur (sur 4000 processeurs) est de l'ordre de plusieurs jours. On peut donc se demander si la DA est vraiment utile dans ce cas. En effet, la DA est plus rapide que la DNS mais pas que la RANS. Cependant, il faut prendre en compte le fait que la DA permet de réduire le temps de calcul de la simulation numérique en permettant d'optimiser les paramètres de l'IBM. De plus, la DA peut être utilisée pour d'autres cas d'étude où la DNS est trop coûteuse.

Pb perso : jusqu'à plusieurs jours d'attente pour lancer une simu.

Tout ce qu'il y a dans le CONES dict Parler des modifs dans le code ici.

- Reduction of the pipe (with the research of the velocity inlet condition)

Toujours mentionner les paramètres utilisés (dans le controlDict et le CONESDict)

2.5 Application of the DA

2.6 Results

///// Reprendre les objectifs définis à la fin de la section 2.1 /////

Comparer les 2-3 résultats : (avant DA, déjà fait section précédente), après simple et "complex" DA

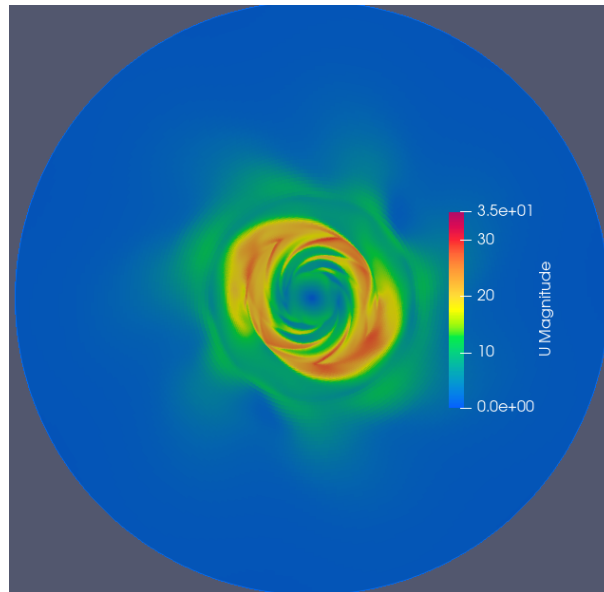


Figure 2.25: Velocity field for $Dx = 4e7$ and $Dy = 2e7$

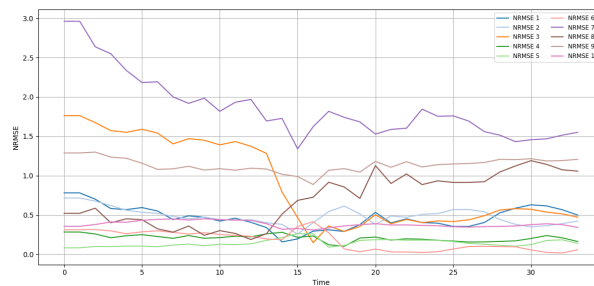


Figure 2.26: Evolution of the NRMSEs over EnKF analysis phases

...

Explication des écarts par différentes choses

Tt ce que j'ai implémenté en plus (déjà détaillé partie précédente normalement)

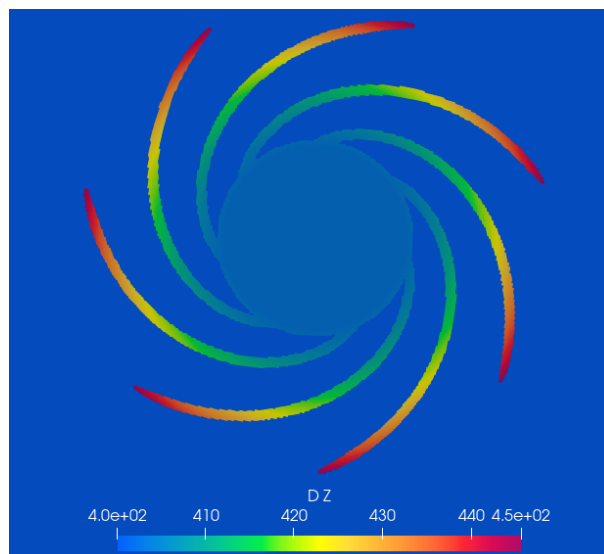


Figure 2.27: Radial expansion of Dz over the impeller radius

Essayer d'améliorer le rendu de l'image ci-dessous en mettant la pompe en transparence

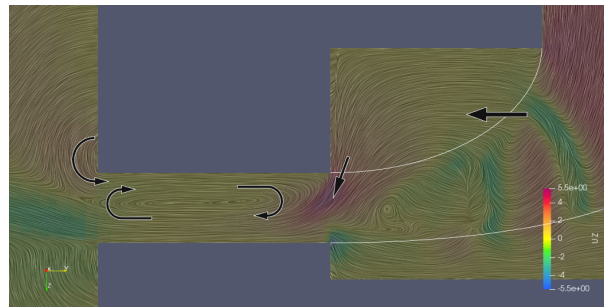


Figure 2.28: Formation of the recirculation bubble in the diffuser

Explication des écarts par la bulle de recirculation.

2.6.1 Conclusion ans perspective

Problème de méthode et pas de miracle ave cette méthode IBM. Interpolation, instationnaire (avantage : on ne contraint plus avec MRF et donc plus facile de réconcilier solution numérique physique et mathématique)

Ouverture : méthode IBM interpolation, article non référencé.

[14] -> ML turbulent channel flow Miguel

"Ouverture / idée" : DA temps réel dans avion impossible pour encore longtemps. Solution : Contrôle actif de la puissance en fonction des mesures de pression dans le compresseur : $f(\text{obs}) \rightarrow$ controle de la puissance Pour ça, se ref au sujet de thèse de Tarane obs pression $\rightarrow$ modèle de reconstruction du champ de vitesse $\rightarrow$ modèle de reconnaissance de patternes de décrochage $\rightarrow$ contrôle de la puissance Peut-être faire deux filtres de Kalman imbriqués ? hyperparamètres type fenetre observation, variance initiale, ...

Voir "TD -> Done et "TDSave -> Done" pour voir tt ce que j'ai fais

The random variations are random but with a reproducibility seed (sed) that allows to have the same result each time the simulation is launched.

//The theorical average radial speed at the input of the diffuser is equal to

$$\frac{Q_{dif_{in}}}{S_{dif_{in}}} = \frac{Q_{pump_{in}}}{2\pi R_3 h_{dif}} = 3.65 m.s^{-1}$$

With $Q_{pump_{in}} = Q_{dif_{in}}$ because the mass flow rate is constant and the flow is incompressible.//
-> No need ??

Parler du cas test profil d'aile de Tom

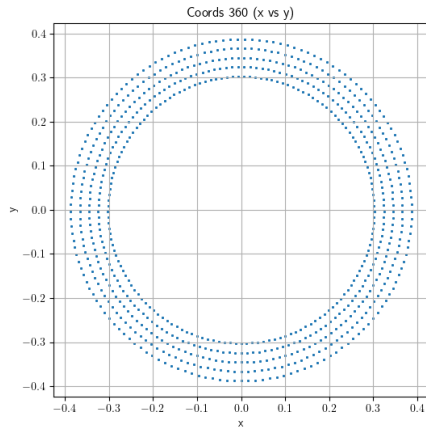


Figure 2.29: Interpolation points over 360 degrees

Methodo (forçage continue (vs interpolation ?)) -> avantages

////Annexe des opérateurs grad, div, nabla, rot, laplacien, laplacien vectoriel, etc...////

The EnKF method will be used to assimilate data from the flow at the observation points and to update the forcing term until it converges to the optimal value. These high-fidelity data are obtained from an experiment explained in the next section 2.3.

//Préparer l'introduction des variables du controlDict et le CONESDict//

Conclusion

The goal of this internship was to optimise the numerical performance of a centrifugal pump using DA. After preparing the experimental data and adapted the OpenFoam solver and CONES code, data was able to be assimilate by the EnKF. First results was not convincing, because a recirculation bubble still present in the diffuser. Few more implementation didn't allow the recirculation bubble to break. Then, ...

Technical and human review

Vérifier les consignes

Outlook

Personal reflection

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Annexes

Annex 1 – Batch code

```
#!/bin/bash                                # To be run with bash
#MSUB -r LCP35                             # Job name
#MSUB -o LCP35_%I.out                      # Output name. %I is the job ID
#MSUB -e LCP35_%I.err                     # Error name. %I is the job ID
#MSUB -q rome                             # Partition
#MSUB -A gen1741                           # Project ID
#MSUB -n 4446                             # Cores nb (35 members x 127 OF) + 1 python
##MSUB -c 2                               # To allocate 2 cores per task (default is 1)
##MSUB -N 1                               # Nb of calcul node (128 cores per node on rome)
#MSUB -T 43200                             # Elapsed time limit in seconds
##MSUB -Q long                             # (upto 3 days)
##MSUB -Q test                             # (short, high priority jobs)
#MSUB -m scratch                           # Execute in scratch directory
#MSUB -@ name@mail.fr:begin,end           # Email address

set -x
cd ${BRIDGE_MSUB_PWD}
module purge
module load gnu/8 mpi/openmpi/4 flavor/python3/cuda python3/3.10.6 openfoam/9
source $FOAM_SOURCE_FILE

./visuNParams.sh                           # Script .sh
ccc_mprun -f app_mySF35.conf               # Run the application with the configuration file
```

Annex 2 – OpenFOAM Directory Tree

```
root
├── 0
│   ├── k
│   ├── omega
│   ├── p
│   └── U
├── 0.orig
│   └── 0
├── system
│   ├── controlDict
│   ├── blockMeshDict
│   ├── decomposeParDict
│   ├── fvSchemes
│   └── fvSolution
├── constant
│   ├── momentumTransport
│   ├── transportProperties
│   └── MRFProperties
├── dummy.foam
├── processor0
├── processor...
└── processorN-1
```

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List of acronyms

AI Artificial Intelligence

BC Boundary Conditions

CERFACS "Centre Européen de Recherche et de Formation Avancée en Calcul Scientifique"

CFD Computation Fluid Dynamics

CFL Courant-Friedrichs-Lewy

CHSCTN "Comité d'Hygiène, de Sécurité et des Conditions de Travail National"

CNRS "Centre National de la Recherche Scientifique"

CONES Coupling OpenFoam with Numerical EnvironmentS

CPER "Contrat de Plan État-Région"

CSE "Comité Social et Économique"

CSR Corporate Social Responsibility

CTI "Commission des Titres d'Ingénieur"

DA Data Assimilation

DNS Direct Numerical Simulation

EKF Extended Kalman Filter

EnKF Ensemble Kalman Filter

ENSAM "École Nationale Supérieure d'Arts et Métiers"

HCERES "Haut Conseil de l'Évaluation de la Recherche et de l'Enseignement Supérieur"

IBM Immersed Boundary Methods

IPSA "Institut Polytechnique des Sciences Avancées"

JFM Journal of Fluid Mechanics

LES Large Eddy Simulation

LMFL "Laboratoire de Mécanique des Fluides de Lille"

Ma Mach number

MPI Message Passing Interface

MRF Multiple Reference Frame

ONERA "Office National d'Études et de Recherche Aéronautiques"

OpenFOAM Open source Field Operation And Manipulation

PDF Probability Density Function

PIV Particle Image Velocimetry

QVT "Qualité de Vie au Travail"

RANS Reynolds-Averaged Navier–Stokes equations

Re Reynolds number

RSE "Responsabilité Sociétale des Entreprises"

RSU "Rapport Social Unique"

SHF "Société Hydrotechnique de France"

SST Shear Stress Transport

U-RANS Unsteady Reynolds-Averaged Navier–Stokes

Résumé Vérifier les consignes

Abstract